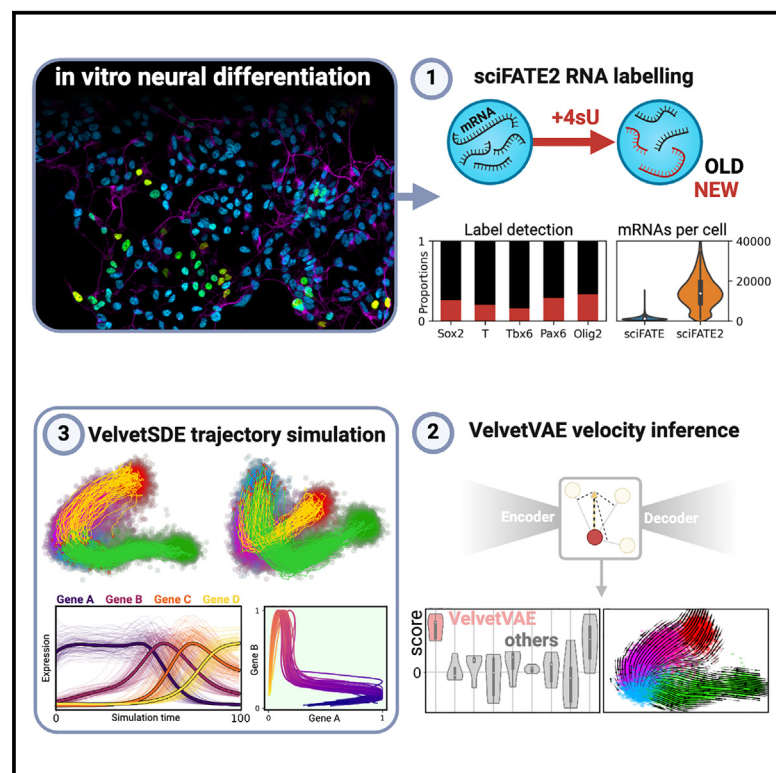


Reconstructing developmental trajectories using latent dynamical systems and time-resolved transcriptomics

Graphical abstract



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In brief

This study presents methods for dynamical modeling from single-cell transcriptomics: an experimental protocol for metabolic labeling that provides increased data quality with single-cell temporal resolution and a generative modeling framework for velocity inference that outperforms other tools and provides trajectory simulations that capture dynamical aspects of cell fate decisions.

Highlights

- Scifate2 is an optimized method for time-resolved single-cell transcriptomics
- VelvetVAE is a deep generative velocity model that outperforms other tools
- VelvetSDE is a model extension for simulating trajectory distributions
- These provide a framework for dynamical modeling from single-cell transcriptomics



Article

Reconstructing developmental trajectories using latent dynamical systems and time-resolved transcriptomics

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SUMMARY

The snapshot nature of single-cell transcriptomics presents a challenge for studying the dynamics of cell fate decisions. Metabolic labeling and splicing can provide temporal information at single-cell level, but current methods have limitations. Here, we present a framework that overcomes these limitations: experimentally, we developed sci-FATE2, an optimized method for metabolic labeling with increased data quality, which we used to profile 45,000 embryonic stem (ES) cells differentiating into neural tube identities. Computationally, we developed a two-stage framework for dynamical modeling: VelvetVAE, a variational autoencoder (VAE) for velocity inference that outperforms all other tools tested, and VelvetSDE, a neural stochastic differential equation (nSDE) framework for simulating trajectory distributions. These recapitulate underlying dataset distributions and capture features such as decision boundaries between alternative fates and fate-specific gene expression. These methods recast single-cell analyses from descriptions of observed data to models of the dynamics that generated them, providing a framework for investigating developmental fate decisions.

INTRODUCTION

The formation of patterned, functional tissues depends on organized cell fate decisions. These are driven by gene regulatory networks (GRNs) that interpret extracellular signals to induce the correct cellular responses.¹ Single-cell RNA sequencing studies have illustrated the complexity of these developmental GRNs, documenting, for example, the temporal patterns of transcription factor (TF) expression in the developing nervous system² and defining sequences of cellular states through development.^{3,4}

A major goal in the field of single-cell genomics has been to model the regulatory mechanisms that underlie observed phenotypes.⁵ However, this remains a difficult task. GRNs are complex and time-dependent, with feedback cycles and emergent behaviors.⁶ To understand such complex processes, dynamical modeling is required.^{7,8}

Single-cell sequencing provides only a static snapshot of cellular composition. Dynamics must be inferred through pseudo-temporal ordering,⁹ which collapses the tens of thousands of data points commonly found in single-cell experiments into an aggregated time series. This can provide valuable insight¹⁰ but lacks the resolution to capture causal relationships between genes (for example, pseudotime information does not improve GRN inference algorithms^{11,12}). Properly capturing the

complex dynamics of developmental gene regulation will be necessary to move from descriptive analyses to quantitative models of causal mechanisms. An important step for this is to capture temporal dynamics at single-cell resolution.

Approaches that resolve time in single cells have been developed: RNA velocity¹³ infers dynamics from the ratio of spliced and unspliced reads. This approach has been widely applied,^{14–18} but it is limited by the technical constraints and confounding factors that arise from relying on gene splicing for temporal information.^{19–21}

An alternative is to use metabolic labeling of RNA with thiouridine (4sU), a strategy that has long been used to study RNA kinetics^{22–24} and that provides a more direct, experimentally controllable measurement of time.^{25–30} In this approach, 4sU incorporates into nascent RNA and is subsequently converted to a cytosine analog with iodoacetamide (IAA) treatment, allowing new RNA to be distinguished from old, based on U-to-C mutations in the sequencing data. Of available methods, only sci-FATE,²⁵ scNT-seq,²⁷ and the more recently published Well-TEMP-seq²⁹ and SLAM-drop-seq³⁰ provide throughput comparable to commercial single-cell platforms, and only sci-FATE can do so without the use of custom microfluidics or microwell devices. However, sci-FATE based on combinatorial indexing is a labor-intensive method with a highly destructive cell treatment protocol that includes liquid nitrogen,



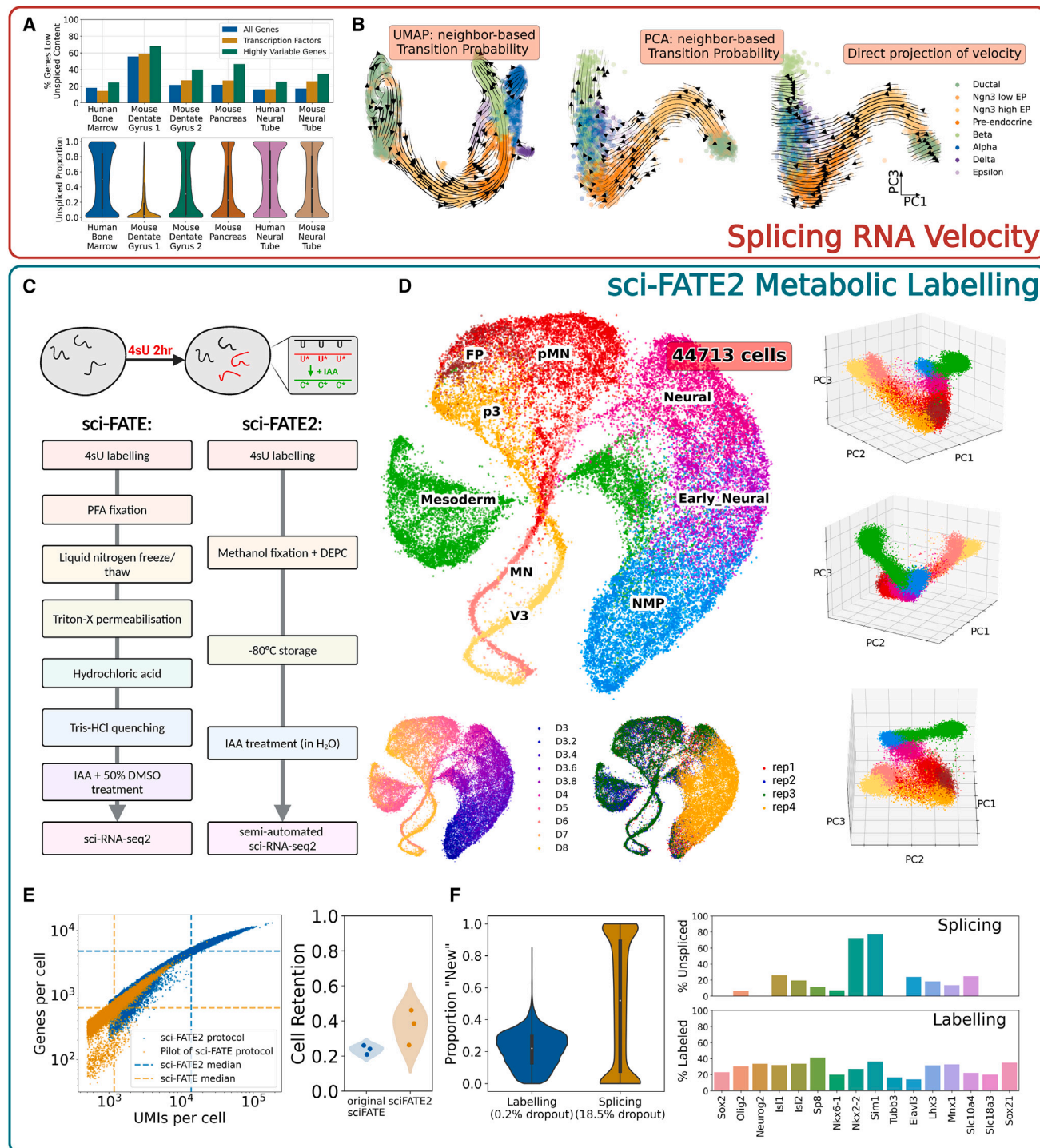


Figure 1. Improved temporal transcriptomics with sci-FATE2

(A) Proportion of genes with low unsplied content (defined as less than 5% total reads being unsplied) (top) and near-uniform distribution of unsplied proportions across genes (bottom) for datasets across tissues (from left to right: Setty et al.,⁴⁸ BioStudies accession S-SUBS8; Hochgerner et al.,⁴⁹ GEO accession GSE95753; La Manno et al.,¹³ GSE104323; Bastidas-Ponce et al.,¹⁵ GSE132188; Rayon et al.,¹⁴ GSE171892; and Delile et al.,² ArrayExpress accession E-MTAB-7320).

(B) Low-dimension visualization of velocity dynamics of pancreas dataset (Bastidas-Ponce et al.,¹⁵ GSE132188), providing representative indication of differences between the neighborhood-based projection heuristic and direct projection; neighborhood-based projection in UMAP, left; neighborhood-based projection in PCA, middle; and directly projected velocities in PCA, right.

(C) Schematic of metabolic labeling protocol and simplified cell treatment protocol in sci-FATE2 compared with sci-FATE.

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paraformaldehyde (PFA), Triton-X, hydrochloric acid, and 50% DMSO. Accordingly, initial pilots of sci-FATE revealed low data quality that would restrict the detection of complex or subtle dynamics.

Alongside experimental methods, a number of computational tools have been developed for inference of gene expression “velocity” from either splicing or labeling data.^{13,20,31–39} In particular, many recent studies have used methods from deep generative modeling^{32,35,36,38,40} to better handle the complex, high-dimensional data. Similar approaches have already been successful in single-cell genomics tasks such as multimodal integration,⁴⁰ perturbation prediction,⁴¹ and data correction.⁴² However, issues remain: first, recent work has found that aspects common across velocity analyses, such as local averaging of data and low-dimensional projection heuristics, can affect the inferred dynamics and lead to inconsistent inferences.^{19–21} Second, velocity inference produces estimates of instantaneous velocity, a single timestep prediction that may be insufficient to accurately capture the long-term dynamics of developmental trajectories. Time-resolved single-cell sequencing methods offer the possibility of transforming single-cell analyses from descriptions of observed data into models of the dynamics that generated them, facilitating a clearer picture of underlying mechanisms. To achieve this, both experimental and computational challenges must be addressed.

Here, we present an integrated framework of experimental and computational methods for modeling dynamical systems of gene expression using time-resolved single-cell RNA sequencing data. Experimentally, we developed sci-FATE2, an optimized, semi-automated version of sci-FATE,²⁵ producing improvements that provide data quality and cellular throughput comparable to commercial platforms.

Computationally, we developed VelvetVAE, a mechanistic variational autoencoder (VAE)⁴³ to model velocity with a lower-dimensional vector field, constraining predicted velocities based on local neighborhood information. To extend our analyses beyond estimation of instantaneous velocity, we developed a model extension, VelvetSDE, that infers global dynamics by embedding the learned vector field in a neural stochastic differential equation (nSDE) system^{44,45} that is trained to produce accurate trajectories that stay within the data distribution.

We show through quantitative benchmarking that VelvetVAE outperforms existing velocity tools across all metrics, without the need for local averaging of data. VelvetSDE’s predicted trajectory distributions map the commitment of cells to specific fates over time and can faithfully conserve known trends while capturing correlative structures between functionally related genes. We use our methods to study neural tube patterning, a complex series of developmental decisions that can be recapitulated *in vitro*.^{46,47} We collect 44,713 time-resolved transcrip-

omes, capturing the differentiation of mouse embryonic stem (ES) cells to neuromesodermal progenitors (NMPs) that then transition to neural and mesodermal identities, with the subsequent specification of neural cell types, floor plate (FP), motor neurons (MNs), and V3 interneurons. Applying our framework to study *in vitro* neural patterning decisions, we describe how this process occurs as two distinct fate decisions and resolve differences in the expression of sonic hedgehog (Shh) regulators that distinguish different neural fates. Our modeling predicts that variable expression of Shh modulators may be involved in signal interpretation and cell fate decision and points toward possible strategies for more precise fate manipulation *in vitro*.

RESULTS

Limitations of current velocity methods

RNA velocity requires that both spliced and unspliced mRNA is detected for each gene. However, splicing varies significantly across genes and many TFs contain no introns. Examining datasets from several tissues from both mouse and human,^{2,13–15,48,49} we see a substantial proportion of genes with minimal unspliced reads (defined as less than 5% of total), and this proportion increases for TFs and highly variable genes (Figure 1A). Often, over a quarter of highly variable genes selected for analysis will have less than 1 in 20 reads classified as unspliced. Given the general low count values observed in single-cell data (for example, the mean non-zero value across the above datasets is 2.4), this suggests that splicing does not provide a robust measurement of nascent transcription for many genes.

Moreover, it has been suggested that the methods used to project velocity into low-dimensional visualizations (such as uniform manifold approximation and projection [UMAP]⁵¹ or t-SNE⁵²) may confound assessment of high-dimensional velocity vectors.^{20,21} In standard RNA velocity analyses, this low-dimensional projection is done using a heuristic based on nearest-neighbor transition probabilities.^{13,53} These projections usually appear to qualitatively capture known biology,^{14–18} but they introduce a confounding factor when attempting to evaluate velocities in full gene expression space. We looked to examine the effect of this projection heuristic on velocity visualization using an example dataset, with velocities inferred using scVelo.⁵³ Projecting velocities into principal-component analysis (PCA) space—either directly or with the standard projection heuristic—revealed substantial differences in the visualized dynamics (Figure 1B). Directly projected velocities appeared less accurate, pointing in directions that are not consistent with the distribution of cells in the dataset. Neighborhood-based projection heuristics may alter velocity vectors by removing erroneous directions that cannot be reconstructed with the distribution of neighboring

(D) Low dimensional visualizations of sci-FATE2 data collected in this study, providing a broad view of cell types, time points, and replicates (D, day; rep, replicate). Left, minimum distortion embedding⁵⁰ of top 5 principal components; right, three-dimensional PCA (colors corresponding to cell types), rotated around the z-dimension corresponding to PC3 ($n = 44,713$).

(E) Comparisons of sci-FATE and sci-FATE2 data quality; left, UMIs and genes per cell—sci-FATE pilot was sequenced with 300M reads and had a 80.9% duplication rate, sci-FATE2 was sequenced with 700 M reads but saw a lower duplication rate of 64.3% (from the same number of inputted cells); $n = 12,075$ and $n = 68,866$, respectively. Right, comparison of cell retention in chemical conversion protocol, $n = 3$.

(F) Comparison of labeling and splicing data for all cells, where new is defined as labeled or unspliced, across all genes (left) and for key genes in neural patterning (right).

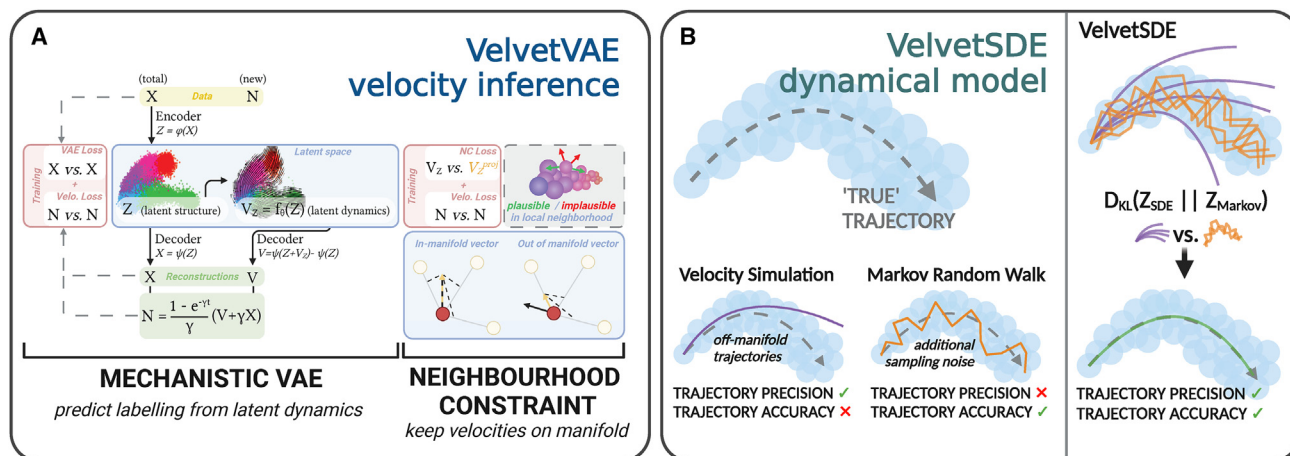


Figure 2. Schematics of velvet models

(A) VelvetVAE is a mechanistic variational autoencoder that learns in a two-stage process: first, learning a latent representation of gene expression, a latent vector field that projects to high-dimensional gene expression and velocity values that can accurately predict labeling information when combined with a biophysical model of transcription. In the second stage, the variational autoencoder is frozen and the vector field's label prediction continues to train alongside a neighborhood constraint that penalizes velocity vectors pointing away from the local neighborhood of observed cells.

(B) VelvetSDE is a deep dynamical systems framework for trajectory prediction. VelvetVAE's vector field is embedded as the drift component of a neural stochastic differential equation (nSDE) framework, and is further trained to maintain trajectories on the data manifold by comparing nSDE simulation distributions with Markov random walk distributions generated from the same velocity dynamics (which necessarily remain on the data manifold).

cells. This suggests that such heuristics potentially conceal inaccuracies in the high-dimensional velocities inferred from splicing data.

sci-FATE2: An optimized metabolic labeling protocol

Observing the inconsistency of splicing data and issues with subsequent velocity inference, we considered metabolic labeling as an alternative, focusing on sci-FATE²⁵ as it offered the best-reported combination of cellular throughput and technical requirements. To test this protocol, we performed a pilot without 4sU labeling but with IAA treatment. This resulted in low data quality, with a median of approximately 1,500 unique molecular identifiers (UMIs) per cell (Figure 1E, left).

To improve data quality, we made several changes to both the IAA treatment protocol (Figure 1C) and the combinatorial indexing protocol (full details in STAR Methods), which provided a substantial increase in detected UMIs per cell (Figures 1E, left, and 2A–2D) and a reduction in cell loss (Figure 1E, right).

With this optimized protocol, sci-FATE2, we capture approximately 12,000 cells per experiment, with a measured doublet rate of 6% (Figure S2E). We observe 15%–20% of reads being detected as labeled (Figure 1F, left). By contrast, in the same data there was a nearly uniform random distribution of unspliced proportions (Figures 1A and 1F), and 18.5% of genes lacked unspliced information, while only 0.2% of genes lacked labeling information (Figure 1F, left). Comparing the proportions of reads classified as “new” by splicing and by labeling for key neural genes, we see that labeling provides a far more consistent signal (Figure 1F, right).

With this optimized protocol, we set out to apply sci-FATE2 to examine the dynamics of patterning and differentiation in the developing neural tube. We used a protocol for directly differentiating mouse ES cells into NMPs, neural progenitors, and spinal cord neurons using 500 nM Shh agonist (SAG) to induce the most

ventral neural tube identities (p3/V3, pMN/MN, and FP).^{46,47} We collected cells daily from day 3 to day 8, and labeled cells with 500 μ M 4sU for 2 h immediately prior to fixation (Figure S1B), having confirmed that this dose led to no observed changes in cell behavior (Figure S4). We collected three full replicates, and a fourth replicate interpolating time points between days 3 and 4 at 5-h intervals, to capture the rapid changes during NMP differentiation. After preprocessing, quality control, and cell-type classification, we retained 44,713 time-resolved transcriptomic profiles that reflected the expected cell types and developmental progressions (Figures 1D and S3).

VelvetVAE: Deep generative velocity inference

To model dynamics, we focus on a generalized idea of velocity that is not specific to splicing dynamics. We applied a biophysical model of RNA transcription, labeling, and degradation as per previous studies^{27,31} (Figure S1A). This model defines a cell's velocity in terms of three observables (labeled reads, total reads, and labeling time) and one free parameter, γ , which represents a gene-specific degradation rate. Although this is more straightforward than the framework required to model velocity from splicing data (with no splicing parameters or latent time required), it remains a difficult task to infer velocity and γ values across thousands of genes.

Thus, to improve velocity inference, we developed a mechanistic VAE framework, VelvetVAE (Figure 2A). Based on previous VAE frameworks for modeling single-cell data,^{54,55} the VAE component learns a latent representation of total observed expression values (default 50 dimensions). Alongside this, a neural vector field embedded within this latent representation predicts latent velocity values. Latent expression and latent velocity are projected through the model decoder and are both fed into the biophysical model of transcription to produce a prediction of labeled expression values that is compared with the

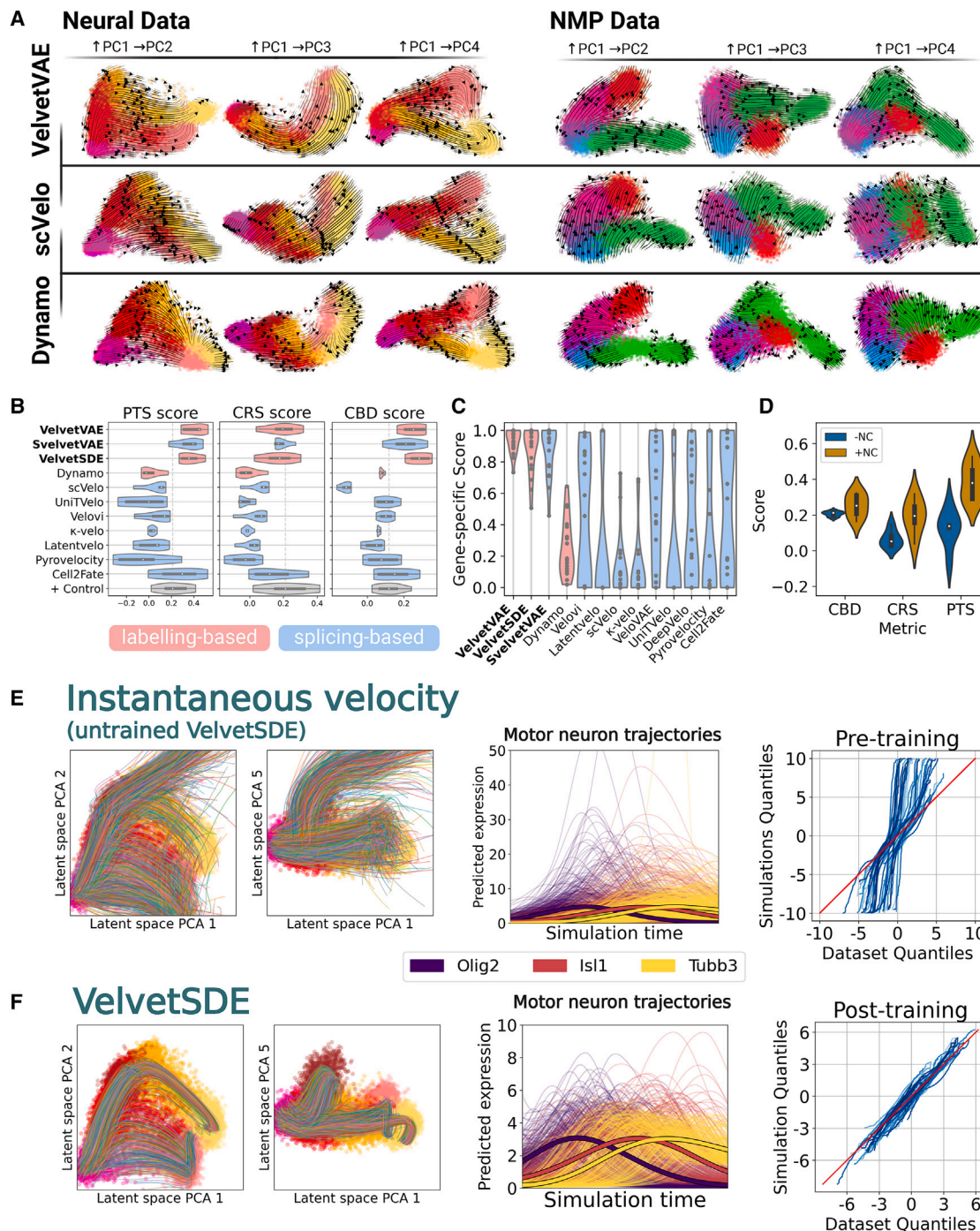


Figure 3. Dynamical modeling of expression dynamics with VelvetVAE and VelvetSDE

(A) Comparison of velocity dynamics inferred by VelvetVAE (top), splicing-based scVelo (middle), and labeling-based Dynamo (bottom) for neural data (left, 20,101 cells) and NMP data (right, 24,262 cells). Velocities are directly projected to PCA space and the top four principal components are visualized.

(B) Quantitative benchmarking of velocity inference tools across six data subsets. Pseudotime score (PTS) and CellRank score (CRS) compare inferred velocities to ground truths constructed with scFates and CellRank. Cross-boundary direction (CBD) compares inferred velocities against user-defined directionalities between clusters. Labeling-based scores in red; splicing-based scores in blue. All velocity inference performed without local averaging of data. Positive controls constructed by comparing ground truths against each other.

(C) Gene-level benchmarking of velocity inference tools. Gene-specific scores compares the sign of inferred velocities against a set of defined genes in defined cell types where the expected sign of velocity is known (for example, Shh velocity is expected to be positive in floor plate cells), $n = 20$ comparisons.

(D) Comparison of VelvetVAE scores with neighborhood constraint (+NC) and without (–NC).

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observed labeling. Thus, the model learns a lower-dimensional representation that can accurately reconstruct the data and a lower-dimensional vector field that can accurately predict the observed labeling dynamics. The degradation parameter, γ , is a latent variable of the model that we assume to be constant to prevent non-identifiability of parameters, which could occur if both γ and predicted velocity could vary across the dataset.

To further constrain the solution space and improve inference, we developed a “neighborhood constraint” that incorporates local neighborhood information (Figure 2A). We reasoned that, in a sufficiently well-sampled developmental trajectory, most cells are expected to be moving toward other observed cells in their local neighborhood (excluding cells at a trajectory’s leading edges that are transitioning to unobserved states). In general, velocity analysis is a form of interpolation, describing transition dynamics between observed states. As such, when a velocity vector points away from all nearby cells in gene expression space, it is more likely to be an erroneous prediction than representative of the underlying biology. VelvetVAE’s neighborhood constraint penalizes velocity predictions that point away from the convex hull of a cell’s nearest neighbors (details in STAR Methods). Coercing the directions in which each cell’s velocity vector can point reduces the solution space for the system’s dynamics in a principled manner. This constraint does not use neighborhood information to average or transform the underlying data, as is commonly done with velocity inference, but simply encodes our expectation that a cell’s velocity vector should, in general, point to other cells in the dataset.

To assess VelvetVAE, we first split the data into two subsets based on known cell fate decisions: NMPs generating either neural or mesodermal cells, and neural progenitor cells differentiating into FP, V3 interneurons, and MNs. We inferred velocity for these decisions separately and visualized velocities in the top four principal components. Comparing velocity dynamics predicted by VelvetVAE to those predicted by two other velocity tools (Dynamo³¹ with labeling, scVelo⁵³ with splicing), we found that VelvetVAE predicted dynamics closely aligned with the expected dynamics of the system, while the other tools did not (Figure 3A).

To quantitatively assess performance, we next split the sciFATE2 data into six subsets for benchmarking (ranging in size from 7,000 to 45,000 cells). Using the trajectory analysis tools scFates⁵⁶ and CellRank,⁵⁷ we developed two parallel “ground truths” pseudotime models (pseudotime score [PTS]; CellRank score [CRS]) constructed to capture the known biological dynamics of each data subset (Figure S4E).

We also used cross-boundary direction (CBD) correctness,³³ which assesses whether velocities correctly predict user-defined directions between clusters. These ground truths test the degree to which VelvetVAE can accurately describe the coarse cell state transitions known to occur in the data.

We compared VelvetVAE to ten previously published velocity tools^{20,31–35,37–39,53} (one using labeling, nine using splicing) and found that VelvetVAE consistently outperformed all other methods across the three metrics (Figure 3B). VelvetVAE performed well with (Figure S9F) and without local averaging of data, a standard prerequisite of other velocity tools. VelvetVAE was the only method to consistently score at or above the scores of positive controls (constructed by scoring ground truths against each other), suggesting that VelvetVAE is performing as well as can be assessed by these ground truths. We found that inclusion of the neighborhood constraint was vital for this strong performance (Figures 3D and S9A).

Using these metrics to compare VelvetVAE against Dynamo³¹ (the only other labeling-based method) on other published metabolic labeling datasets with time-course information,^{25,29,30} we see VelvetVAE scores higher in this context as well (Figure S5D). However, limitations in dataset sizes, time-course durations, and the presence of confounding technical noise make these results more difficult to interpret (Figures S5A–S5C) and highlight the importance of the experimental refinements introduced in sciFATE2.

The above benchmarking assesses whether velocities correspond with expected cellular dynamics. We next sought to assess VelvetVAE’s ability to correctly predict gene-level dynamics. For this, we developed a “gene-specific score,” where the direction of predicted velocities for a specific gene in a specific cell type are compared against expectation (for example, *lrx3* velocity is expected to be positive in early neural cells; *Shh* expression is expected to be positive in FP). Again, we found that VelvetVAE outperformed all other tools by this measure (Figure 3C).

Given VelvetVAE’s strong performance with labeling data, we next asked whether a similar approach would work with splicing information. We built a modified version of the model, SvelvetVAE, that handles splicing data (details in STAR Methods), and found that this model also outperformed other tools (Figures 3B and 3C). However, we found that inference from splicing data was negatively affected by performing local averaging in preprocessing (Figure S9F). Testing SvelvetVAE with six publicly available datasets,^{2,3,13,15,49} we found that it modestly outperformed other tools (Figure S6A).

However, we noted that all tools performed relatively poorly and further examination revealed that even the best-scoring model instance (in this case, UniTelo³³ with dentate gyrus data⁴⁹) failed to fully capture the expected dynamics (Figures S6B–S6D). Moreover, we found that SvelvetVAE inference from splicing data is sensitive to hyperparameters (Figures S6D and S6E) in a way that is not observed when modeling from labeling data. Overall, the performance of SvelvetVAE points toward the general utility of our method and, in particular, the neighborhood constraint, but suggest

(E) Trajectory simulation of early neural cells in neural dataset from VelvetVAE dynamics without further training. Left: each line represents a different cell’s trajectory. Trajectories visualized with PCA of VelvetVAE latent space. PCA1 vs. PCA2 displays neural trajectories, while PCA1 vs. PCA5 additionally captures floor plate trajectories. Middle-right: projecting trajectories through the model decoder produces gene expression values for key genes in motor neuron trajectories that are noisy and implausible. Right: quantile-quantile plot (Q-Q plot) shows low correspondence between latent distribution of simulation and latent distribution of dataset. Each line represents a different latent dimension of model latent space.

(F) The same plots, after VelvetSDE training. Left: trajectories stay within data manifold. Middle-right: gene expression profiles are more realistic. Right: simulation and dataset distributions show close correspondence.

that this method will not remove or resolve confounding factors known to be present in the splicing information of different datasets.^{19,58,59} More generally, the benchmarking scores from publicly available splicing data suggest that, alongside robust velocity tools, high-quality temporal information (such as that provided by metabolic labeling) is required for reliable inference of dynamics.

VelvetSDE: Dynamical modeling with neural SDEs

We next reasoned that we could use numerical integration of VelvetVAE's vector field to simulate each cell's developmental trajectory. However, we found that the resultant trajectories rapidly deviated from the data manifold (Figure 3E, left). Moreover, gene expression profiles obtained by projecting trajectories through the model's decoder exhibited implausible and highly variable values (Figure 3E, second right), while the latent distribution of predicted trajectories did not correspond well with the latent distribution of the underlying data (Figure 3E, right).

We hypothesized that this issue could result from the fact that VelvetVAE models dynamics over a single timestep only. As such, prediction errors that might be minor in the context of a single timestep may accumulate across successive timesteps of a trajectory to produce major systematic errors. We thus reasoned that long-term dynamics of trajectories must be explicitly modeled.

A common way to do this is to use velocity to construct a transition matrix describing the probability of each cell transitioning to each other cell in the dataset.^{53,57} Cell trajectories can then be simulated as a series of transitions, or jumps from cell to cell, along the data manifold, with each jump being determined by probabilities in the velocity-derived transition matrix. This approach draws upon the fact that random walk operators on a graph (which the above process describes) can approximate the dynamics of an underlying stochastic process.^{9,57,60} However, because dynamics are approximated by jumps between finitely sampled cell states, this method can introduce random error.¹⁹ With our data and inferred velocity dynamics, we observed Markov random walk simulations to be highly noisy, producing trajectories that did not clearly delineate between, or cluster into, the expected biological cell fates (Figure S7).

To summarize: simulation through numerical integration produced continuous trajectories that were accurate over short timescales but suffered from systematic errors over longer timescales. Simulation through Markov random walks produced trajectories that were heavily affected by random sampling noise, but provided better accuracy over longer timescales (a result of trajectories being restricted to the data manifold).

We thus explored whether a “best of both worlds” approach was possible: whether it was possible to achieve smooth trajectories with long-term accuracy by training VelvetVAE's trajectories against those produced by Markov random walks.

To this end, we developed VelvetSDE, an extension to our model that uses an nSDE framework^{44,45} to model trajectory dynamics as a drift-diffusion system (Figure 2B). In this framework, the SDE is a neural network, and numerical integration of the system is compatible with back-propagation and gradient descent, allowing the neural network to be updated based on the results of trajectory simulations. The drift component of VelvetSDE is

the neural vector field trained from VelvetVAE. The diffusion component is set as scalar diagonal Brownian motion (i.e., a simple random walk) with magnitude that is set to approximate the noise magnitude observed in Markov random walk simulations for training. To train VelvetSDE, nSDE trajectory distributions and Markov random walk distributions are produced for each cell, and the model's loss is defined as the Kullback-Leibler divergence of nSDE trajectory distributions from Markov trajectory distributions (assuming Gaussian distribution across each timestep of a simulation's trajectories). This training step is designed to correct for the systematic errors observed in numerical integration of VelvetVAE's vector field. By using a stochastic framework, we can approximate the noise magnitude of random Markov walk with the model's own diffusion component, allowing the model's drift component to be compared with the underlying drift of the Markov random walks. From another perspective, this stochastic approach prevents the model from overfitting to the sampling noise present in Markov random walks. Once trained, VelvetSDE's noise component can be adjusted or removed, allowing trajectory simulation without the large sampling noise of Markov random walks.

After training VelvetSDE, simulated trajectories successfully stayed within the data manifold (Figure 3F, left). VelvetSDE's velocities still perform well in quantitative benchmarking, suggesting that biological dynamics are conserved (Figure 3B). Predicted expression values were no longer implausible (Figure 3F, second right) and latent distribution of simulations show concordance with data (Figures 3F, right, S10D, and S10E). These results indicate that VelvetSDE can simulate trajectories that faithfully recreate the distributions of the underlying data.

Predicting cell fate dynamics in NMPs

To explore VelvetSDE's predicted trajectories, we first focused on cell fate decisions in NMPs. Producing deterministic predictions (i.e., by setting the nSDE's noise size to zero) for all day 3.2 (day 3 + 5 h) cells, we found trajectories clustered into two distinct fates: neural and mesodermal (Figure 4A, left). By adding noise to the simulation (choosing a noise size of 0.1, based on visual inspection, Figure S10B), we could then assess which cells had uncertain fates, such that their predicted fate would vary over multiple stochastic simulations. We simulated each cell's trajectory 10 times, classifying cells as “neural fate only,” “mesoderm fate only,” or “mixed-fate” (Figure 4B). PCA visualization of these classifications in VelvetSDE's latent space revealed a clear “decision boundary” between the two fate-committed regions (Figure 4C). We found this boundary, which reflects the point of fate commitment, to align with the boundary between expression and velocity values of known neural, NMP, and mesodermal marker genes (Figure 4E), indicating that this region in gene expression space represents bipotent NMPs.

We examined how this predicted mixed-fate NMP population evolves over time. The proportion of mixed-fate cells decreased monotonically until none were left by day 5 (Figure 4C). To examine the effect of VelvetSDE's noise on this mixed-fate population, we repeated trajectory simulation across a range of noise magnitudes and observed that, while increasing noise magnitude increased the proportion of cells with mixed fates (as would be expected), the mixed-fate cell population had largely disappeared by day 4 (Figure 4D) across noise

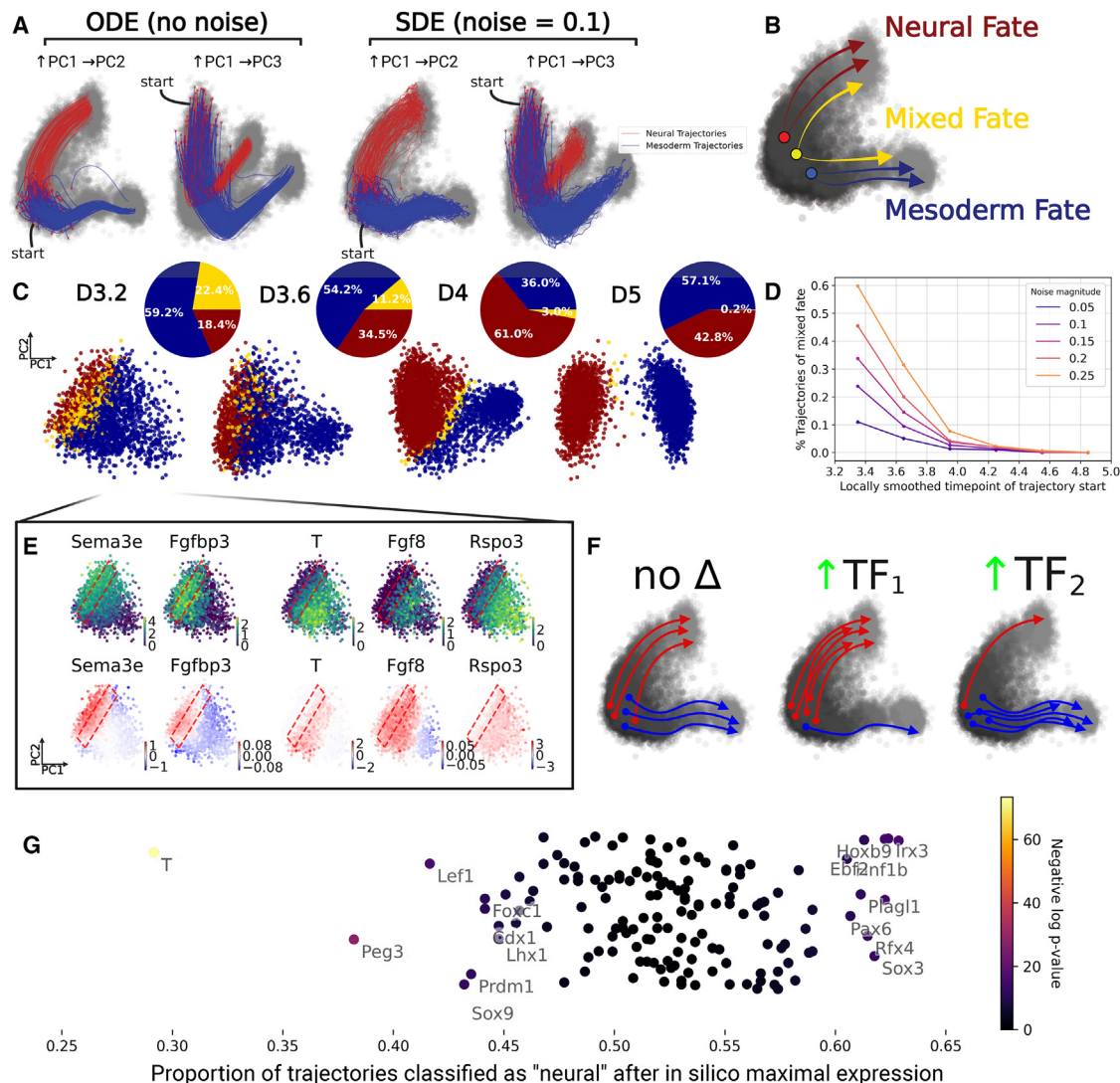


Figure 4. VelvetSDE analysis of cell fate decision boundaries

(A) Simulating trajectories from day 3.2 NMPs (labeled "start") in NMP dataset (24,262 cells) without (left) and with (right) stochastic noise (noise size of 0.1, empirically chosen). Trajectories cluster into two distinct fates, neural (red) and mesodermal (blue).

(B) Simulating multiple stochastic trajectories from a single cell can give three results: trajectories that are entirely neural (red), entirely mesodermal (blue), or a mixture of the two (yellow).

(C) Simulating 10 stochastic trajectories per cell and classifying cells based on the predicted fates as neural, mesodermal, or mixed. Cells visualized in PCA of latent representation. Pie charts represent proportions of each group at each time point.

(D) Trajectory simulation and mixed-fate classification across different time windows, with varying noise sizes.

(E) Expression (top, color represents $\log(x + 1)$ transformed expression values) and velocity (bottom, color represents predicted velocity value, linear scale) of neural, NMP, and mesodermal markers in day 3.2 NMPs, visualized as PCA of cells in latent space. Red box highlights the decision boundary region of mixed-fate cells.

(F) By altering the expression of different transcription factors and examining the resultant change in trajectory distributions, we can examine which genes contribute to the decision boundary geometry in high-dimensional gene space.

(G) Maximal expression alteration experiments across all transcription factors (TFs). Genes plotted based on the proportion of trajectories classified as neural after maximal expression alterations and colored by negative log p value, calculated with a binomial test. $n = 641$ mixed-fate cells from day 3.2.

magnitudes, consistent with the observation that neural commitment genes such as *Olig2* are induced on day 4.⁴⁷

To further validate the model's predictions, we next looked at which genes contributed to the decision boundary, focusing on TFs (Figure 4F). To do this, we set expression for a given TF to the maximum observed value across all mixed-fate cells and

re-projected these altered data to the VelvetSDE latent space. We then simulated trajectories from altered data points to observe how increasing each gene's expression values affected the distribution of predicted fates. By doing this, we aimed to find the genes that most highly corresponded with the geometry of the decision boundary dimensions in gene expression space.

Ranking genes by the magnitude of effect, we see that the highest-ranked hits for increasing or decreasing the “neural” proportion of fates include many TFs known to play a role in neural or mesodermal specification, respectively (Figure 4G), confirming that the axes of the decision boundary corresponds with expected biology. Visualizing the loading contribution of these highly ranked genes to the model’s latent space, we see broad correspondence to the observed neural-mesodermal decision directions (Figure S11C).

In summary, stochastic trajectory simulations can highlight fate commitment points and identify factors that may be involved in fate specification. This approach provides a highly resolved picture of the structure and timing of cell fate decisions, facilitating the generation of hypotheses regarding fate commitment, potency, and reprogramming that can be subsequently tested through genetic perturbation and lineage tracing experiments.

Examining the dynamics of *in vitro* neural patterning

We next focused on neural patterning trajectories. Simulating all “early neural” cells (without added noise) revealed three fate clusters: MN, V3 interneuron (V3), and FP (Figure 5A). By projecting trajectories through the model’s decoder, we could generate gene expression time-series distributions that show key differences between fates, while displaying considerable variability across the individual time series of a given fate.

To test whether biological information is conserved or removed by this variability, we focused on MN trajectories. We first defined ordered pairs of expression that are expected to be consistent across all trajectories (for example, Olig2 expression should precede Neurog2 expression⁴⁷). For each ordered pair, we calculated the percentage of trajectories that correctly have the time of maximum expression ($t_{\max}$) for the first gene before that of the second gene. We compared VelvetSDE with and without noise to baseline models of either a Markov random walk based on velocity dynamics or randomly sampling from binned pseudotime values calculated with scVelo.⁵³ With no noise, VelvetSDE simulations almost perfectly conserve expected gene orderings (Figure 5B), outperforming both baselines. As would be expected, the score of VelvetSDE decreased as noise size increased (Figures 5B and S12C), scoring comparably to pseudotime binning with noise magnitude 0.1 (Figure 5B) and comparably to Markov random walks with noise magnitude of 0.2 (Figure S12C). Visualizing the distribution of $t_{\max}$ values for related pairs of genes shows strong correlations that are less evident for other unrelated genes (Figure S12E). This suggests that the correlative structure contained in trajectory distributions could help distinguish stably related genes from other unrelated genes, although more work is required to establish how biologically meaningful these correlative structures are.

Having established the strong conservation of biological information in VelvetSDE trajectories, we next investigated the dynamics of neural fate decisions in the *in vitro* system. *In vivo*, the three fates observed in our data (MN, V3, FP) are exposed to different concentrations and durations of Shh signaling.⁶¹ However, *in vitro*, all cells are exposed to the same concentration and timing of SAG treatment (500 nM daily from day 3). This raises the question of how these distinct fates arise in a homogeneous signaling environment; that is, whether different cells

interpret signals differently from the start (i.e., parallel fate selection) or whether fates appear sequentially due to prolonged signal exposure (i.e., sequential fate selection). We reasoned that with the increased dynamical resolution of VelvetSDE simulations, we might be able to provide insight into the dynamics of these patterning decisions.

Having observed that MNs arise before V3 interneurons (consistent with previous studies⁶²), we noted that Olig2 expression was detectable in V3 and FP trajectories, but at a lower level than in MN trajectories (Figure 5B), suggesting that high levels of Olig2 expression are associated with commitment to MN fate (consistent with previous work⁴⁷).

We next examined when trajectories split from each other by visualizing cross-section clustermaps of all trajectories, clustered by their eventual fate. Doing so at different simulation time-steps, we find that there is initially very little to distinguish the three fates (Figure 5D, left). By an early stage (20 timesteps of 100), the gene expression profiles of MN trajectories were distinguishable from V3 and FP trajectories, but these latter two fates were less distinct (Figure 5D, middle). Subsequently, the three fates could be clearly distinguished (Figure 5D, right).

Examining median expression time-series plots of key genes for the three different fates indicated that many early neural markers, such as Pax6 and Irx3, have consistent expression profiles regardless of fate (Figure 5E, early genes), suggesting that the exit from an early neural state is similar across the three fates.

Subsequent neural progenitor markers showed clear distinctions between fates, starting with Olig2 and Nkx2-9 distinguishing MN from non-MN fates (Figure 5E, progenitor genes), consistent with the role of Nkx2-9 in the specification of both FP and V3 cell types.^{63,64} Of note, while Olig2 and Nkx2-9 appear to be the earliest marker genes to separate MN from the other fates, the earliest distinction between V3 and FP fates appears to be the induction of Shh itself in FP (Figure 5E, Shh genes).

Additional to the expected variation in TF expression, we also observed fate-specific expression patterns for modulators of Shh signaling: expression of Hhip (a negative regulator of Shh⁶⁵) appears higher in MN trajectories, while expression of Boc (a Shh co-receptor⁶⁵) is higher in FP and V3 fates (Figure 5E, Shh genes), with similar trends observed in the gene expression data (Figure S11E).

Visualizing time series as gene-gene plots confirms the distinct expression patterns of key genes in V3 vs. MN trajectories: known antagonists Nkx2-2 and Olig2 display distinct phase portraits in the two fates, a pattern that is partially recapitulated in the Boc-Hhip phase portrait, while for Hhip and Olig2, the phase portrait shows co-expression that is considerably stronger in MN trajectories (Figure 5F).

The dynamics described by this analysis are consistent with a two-stage decision: first, MN vs. FP/V3, followed by a V3 vs. FP decision. This is in line with previous findings regarding the timing of differentiation *in vivo*⁶⁶ and *in vitro*⁴⁷ and corresponds with recent findings that chromatin landscape changes distinguish FP and V3 cells from other neural progenitors.⁶⁷ The time delay between MN fates and V3/FP fates may correspond to the additional duration of SAG necessary to direct cells that are still Shh-responsive (i.e., cells that have not yet expressed sufficient Olig2 to differentiate into MNs) to shift from an otherwise-expected MN fate to a V3/FP fate.

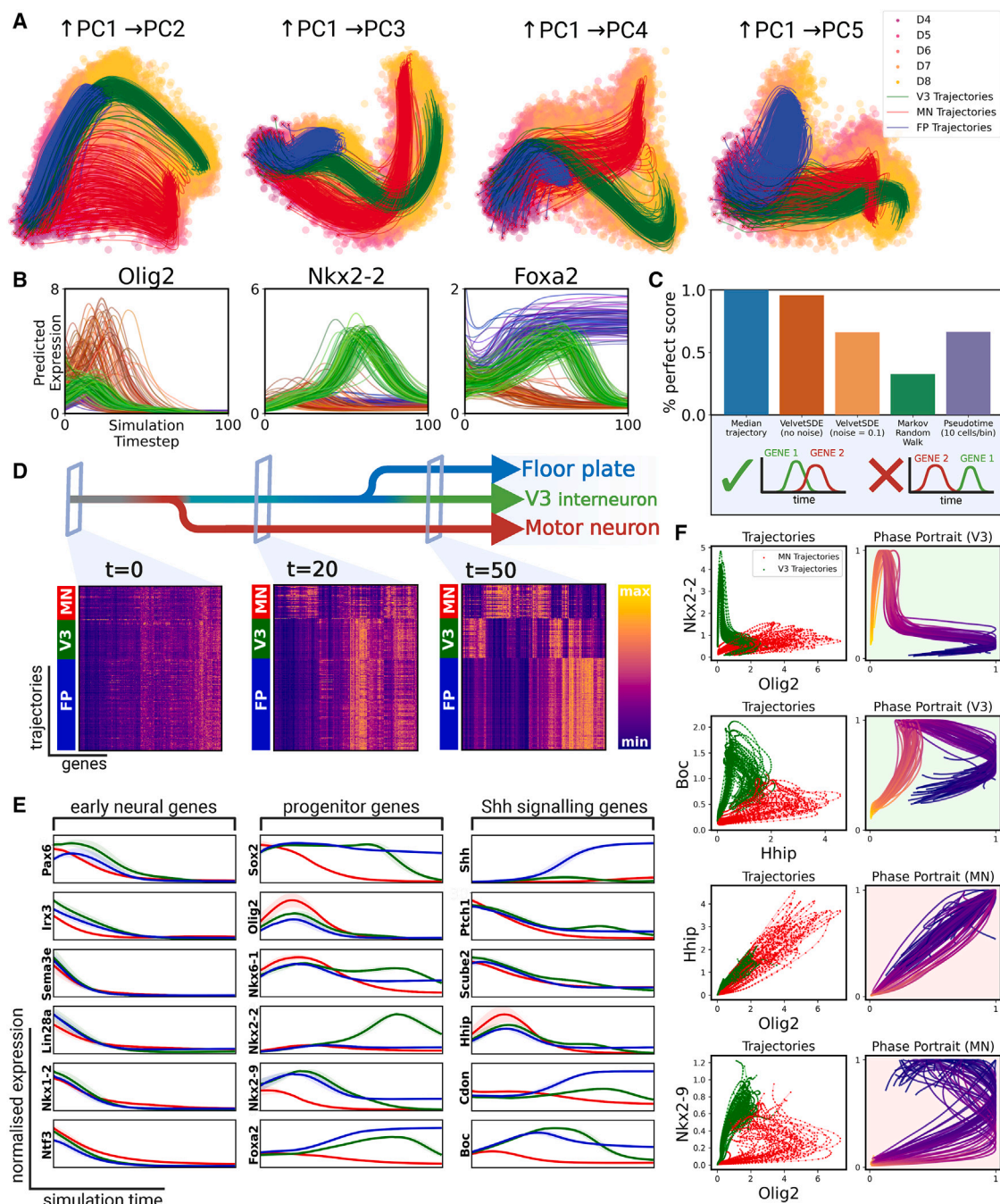


Figure 5. VelvetSDE analysis of neural trajectories

(A) Simulated trajectories from 3,329 early neural cells in neural dataset, clustered into motor neuron (MN; red), V3 interneuron (V3; green), and floor plate (FP; blue) fates, and visualized in first five principal components of PCA.

(B) Projecting trajectories back to gene expression space give a distribution of time series that reflect expected dynamics in key genes.

(C) Scoring trajectories on their ability to conserve expected orderings of defined gene pairs (e.g., Olig2 before Neurog2 in motor neuron trajectories), providing a comparison of VelvetSDE (with and without a noise level of 0.1), we can compare to baselines: the median predicted trajectory, Markov random walk simulations, or binning cells based on velocity-informed pseudotime values. Score represents the percentage of trajectories that perfectly capture all defined orderings.

(D) Cross-sectional clustermaps of all trajectories at different timesteps along the simulation. Genes are clustered, trajectories are grouped by their eventual fate. Schematic above represents the observed dynamics of MN cells splitting first, before V3 and FP cells split from each other. Color represents predicted expression normalized across cells.

(E) Median time series for each fate, showing early neural genes (left), progenitor markers (middle), and sonic hedgehog (Shh) signaling genes (right) for the three trajectories: MN (red), V3 (green), and FP (blue). Confidence interval ranges calculated from each replicate's median trajectories ($n = 3$).

(F) Phase trajectories (left) and phase portrait plots (right) for key pairs of genes in V3 and MN trajectories. Phase portraits move from blue to yellow through time and are produced by size-normalizing each trajectory along both axes of the gene-gene plot.

However, neural fate decisions do not appear to be entirely sequential or solely down to Olig2 levels, with Nkx2-9 and Shh regulators Boc and Hhip also showing early differences in expression between fates. This observation is consistent with previous findings documenting the complex differential regulation of Shh co-receptors such as Boc,⁶⁸ and suggests a more complex picture of fate determination than a purely transcription-factor-mediated regulatory network. However, it is not clear whether the observed variation in Shh modulators is a cause or consequence of TF expression, and more work is needed to understand the causal relationships between signal modulation and signal interpretation by GRNs. More generally, these findings raise the possibility that specific targeting of Shh regulators such as Hhip and Boc could provide additional means to control the course of neural fate decisions *in vitro*.

DISCUSSION

Here, we have developed an integrated framework of experimental and computational methods for dynamical modeling of gene expression from time-resolved transcriptomics data. Although splicing and metabolic labeling data are both regularly used for the inference of single-cell temporal dynamics, we found that the signal quality of splicing data is insufficient, while the data quality of a recently developed metabolic labeling protocol was low. Our optimized, semi-automated protocol, sci-FATE2, provides a controllable, consistent temporal signal with data throughput and quality comparable to commercial platforms.

To infer dynamics with sci-FATE2 data, we developed a velocity inference method, VelvetVAE, which builds local neighborhood information into the inference process of a mechanistic, generative model, resulting in velocity predictions that are more accurate and score consistently higher than existing methods across quantitative benchmarking. Our model extension, VelvetSDE, is further trained to correct systematic errors in trajectory prediction. With this, we can simulate trajectories of individual cells to produce trajectory distributions that conserve known biology and recreate the distributions of the underlying data. Applying the approach to the question of a cell fate specification *in vitro*, we describe the dynamics of sequential fate decision making and resolve fate-specific expression differences in regulators of Shh signaling.

Future work can extend the capabilities of the experimental methods presented: the throughput of sci-FATE2 can be increased with additional rounds of indexing,⁶⁹ while further extensions of labeling (e.g., additional thio-nucleotides,⁷⁰ variable labeling durations) could offer further improvements to data quality and temporal signal resolution.

VelvetSDE's stochastic framework also opens up the possibility for future work to learn more sophisticated and biologically motivated models of noise and uncertainty, allowing analyses that robustly handle stochasticity as a meaningful part of the biological systems we study. This framework also provides a natural route toward implementations that integrate metabolic labeling information alongside longer-term information such as lineage tracing.

Through the integration of temporal transcriptomics and mechanistic generative modeling, we can capture high-quality

information on the dynamics of expression. This allows dynamical analysis of cell fate decisions and provides a framework for modeling complex regulatory mechanisms in development.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cels.2024.04.004>.

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AUTHOR CONTRIBUTIONS

R.J.M. and J.B. conceived the project, interpreted data, and wrote the manuscript. R.J.M. designed and performed experiments, developed the modeling framework, and performed computational and data analysis. R.J.M. and D.M.S. developed and optimized automation of the experimental protocol.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
LIF	Merck	ESG1107
CHIR99021	Axon Medchem	1386
bFGF	Peprotech	100-18B
Retinoic Acid	Merck	R2625
DAPI	Thermo Scientific	D1306
SAG	Calbiochem	566660
4sU	Sigma	T4509
DEPC	Sigma	D5758
SuperScript IV reverse transcriptase	Thermo Scientific	18090010
NEBNext® High-Fidelity 2X PCR Master Mix	NEB	M0541L
NEBNext Ultra™ II Non-Directional RNA Second Strand Synthesis Module	NEB	E6111L
Critical commercial assays		
Kapa mRNA Hyperprep with polyA capture beads	Kapa	KK8421
PowerUp SYBR Green Master Mix	Thermo Scientific	A25742
Deposited data		
Raw and processed sequencing data	This paper	GEO: GSE236520.
Bulk RNA-seq differential expression data	This paper	Zenodo, https://doi.org/10.5281/zenodo.10854059
mRNA half life data	Sharova et al. ⁷¹	N/A
qPCR data	This paper	Table S2
Experimental models: Cell lines		
HM1 mouse embryonic stem cells	Thermo Scientific	N/A
Oligonucleotides		
Full list of primer oligos in Table S1	Integrated DNA Technologies	N/A
Tn5-N7: GTCTCGTGGGCTCGGA GATGTGTATAAGAGAC	Integrated DNA Technologies	N/A
Tn5-ME: [phos]CTGTCTCTTATACACATCT	Integrated DNA Technologies	N/A
Software and algorithms		
VelvetVAE analysis package	This paper	https://github.com/rorymaizels/velvetVAE (https://doi.org/10.5281/zenodo.10815618)
Scanpy	Wolf et al. ⁷²	https://scanpy.readthedocs.io/en/latest/index.html
Pytorch	Pytorch	https://pytorch.org/docs/stable/
torchsde	Kidger et al. ⁴⁴	https://github.com/google-research/torchsde
Scvelo	Bergen et al. ⁵⁹	https://scvelo.readthedocs.io/en/stable/
Dynamo	Qiu et al. ³¹	https://dynamo-release.readthedocs.io/en/latest/
Velovi	Gayoso et al. ³⁴	https://velovi.readthedocs.io/en/latest/
Unitvelo	Gao et al. ³³	https://unitvelo.readthedocs.io/en/latest/
kvelo	Marot-Lassauzaie et al. ²⁰	https://github.com/HaghverdiLab/velocity_package
pyrovelocity	Qin et al. ³⁷	https://github.com/pinellolab/pyrovelocity/tree/main
Latentvelo	Farrell et al. ³⁵	https://github.com/Spencerfar/LatentVelo/blob/main/examples/Synthetic%20bifurcation.ipynb
Cell2fate	Aivazidis et al. ³⁸	https://github.com/BayraktarLab/cell2fate
Velovae	Gu et al. ³²	https://github.com/welch-lab/VeloVAE

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
deepvelo	Cui et al. ³⁹	https://github.com/bowang-lab/DeepVelo/blob/main/deepvelo/model/model.py
dynast	Qiu et al. ²⁷	https://dynast-release.readthedocs.io/en/latest/
DESeq2	Love et al. ⁷³	https://bioconductor.org/packages/release/bioc/html/DESeq2.html
sciFATE2_processing	This paper	https://github.com/rorymaizels/sciFATE2_processing (https://doi.org/10.5281/zenodo.10815622)
Code for analysis	This paper	https://github.com/rorymaizels/Maizels2023aa (https://doi.org/10.5281/zenodo.10815620)
Other		
Mosquito liquid handler	SPT Labtech	https://www.sptlabtech.com/products/mosquito

RESOURCE AVAILABILITY

Lead contact

Further information and requests for reagents may be directed to, and will be fulfilled by, the lead contact, James Briscoe (james.briscoe@crick.ac.uk).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- Raw sequencing data and processed sequencing data can be accessed with the GEO accession: GSE236520 and are publicly available as of the date of publication. RNA-seq differential expression data and model benchmarking data are available on Zenodo with [<https://doi.org/10.5281/zenodo.10854059>]. Microarray half-life data is available from Sharova et al.⁷¹ RT-qPCR data is available as a supplementary file.
- All original code for raw data processing and label detection with dynast is available at: https://github.com/rorymaizels/sciFATE2_processing. All original code for computational analysis is available at: <https://github.com/rorymaizels/Maizels2023aa>. The deep learning framework, available as the software package, velvetvae, can be installed from: <https://github.com/rorymaizels/velvetvae>.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell Line

The mouse ES HM1 line was used for all experiments. ES cells were maintained at 37°C with 5% carbon dioxide (CO₂) and routinely tested for mycoplasma.

METHOD DETAILS

Cell culture and differentiation

Cells were cultured and differentiated as described previously.^{46,47,67} HM1 (Thermo scientific) ES cells were maintained in ES cell medium with 1,000U/ml LIF (Merck) on mouse embryonic feeder cells (mitotically inactivated). These cells were dissociated in 0.05% Trypsin (Gibco) and plated on tissue culture plates for 20 minutes twice successively to remove feeder cells. Remaining supernatant cells were then plated on CellBind six-well plates (Corning) pre-coated with 0.1% gelatine solution (coated overnight at 37°C), 60,000 cells per well in 1.5ml N2B27 + 10ng/ml bFGF. On day 2, the medium was replaced with N2B27 with 10ng/ml bFGF and 5μM CHIR99021 (Axon) for 20 hours. Subsequently and every 24 hours afterwards, medium was replaced with N2B27 with 100nM RA (Sigma) and 500nM SAG (Calbiochem).

Cell labelling and treatment

Cells were incubated in 500μM 4sU (Sigma, made into 500mM stock dissolved in DMSO) for two hours in the appropriate medium in the dark. Collection was then done as swiftly as possible to minimise the time between labelling and fixation. For this, cells were washed in PBS, dissociated in 500μl room temperature accutase for two minutes, spun at 1000g for 3 minutes. For fixation, cells were resuspended in 400μl PBS + 0.1% DEPC (Sigma) + 10mM DTT (Sigma) and 1600μl Methanol + 0.1% DEPC + 10mM DTT

was added slowly, drop-wise to cells. Cells were kept on ice, rocking at 20rpm for 30 minutes, and were then stored at -80°C for up to three weeks. Note that the melting point of methanol is -97.6°C , and as such this fixation protocol will not result in freeze- or thaw-associated damage to cells. The number of wells used for a single fixed sample was chosen such that 5–10 million cells were fixed per sample.

sci-FATE2

Summary of main protocol changes: Noting that PFA fixation, Triton-X and freeze-thaw cycles are all associated with increased RNA damage or loss,^{74,75} we used methanol fixation and liquid-phase storage at -80°C . Finding that enzymatic RNase inhibitors are ineffective,⁷⁶ we add DEPC to our fixation and wash buffers. To eliminate manual errors and reduce time spent with samples on ice, we automated reverse transcription and PCR steps with the Mosquito HV (SPT Labtech). To minimise over-amplification, we reduced the number of PCR cycles from twenty to ten, and concentrated the library with SPRI beads. To reduce the harshness of the IAA chemical conversion step, we removed the use of hydrochloric acid and 50% DMSO, finding that doing so did not negatively impact the detection of labelled reads or the total number of UMIs per cell (Figures S2A and S2B), and provided an increase in cellular retention through the protocol (Figure 1E, right).

For chemical conversion, cells were kept on ice at all times and all spins were done at 2000g, 4°C for 5 minutes. Cells were first moved from -80°C onto ice for 3 minutes and gently resuspended. Cells were then spun down and resuspended in 1ml PBS + 0.1% DEPC + 3% v/v dissolved BSA solution (NEB) + 10mM DTT, spun again and resuspended in 100 μl PBS + 3% v/v dissolved BSA solution. To this suspension, 220 μl water, then 40 μl sodium phosphate buffer (0.5M, Thermo Scientific) then 40 μl 100mM IAA (Sigma, dissolved in ethanol) were added. Cells were incubated at 50°C for 15 minutes, being gently resuspended every five minutes. Cells were then added to a quenching mix of 1.5ml PBS + 3% v/v dissolved BSA solution and 5mM DTT. Cells were spun down and resuspended in PBS at a concentration of 1 million cells per ml. Sci-FATE2 library preparation was performed immediately. Cells were distributed 2 μl per well into a LoBind 384-well plate (Eppendorf) using Mosquito HV (SPT Labtech). 1 μl of 25 μM oligo-dT primer (5'-ACGACGCTCTCCGATCTNNNNNNNN[10bp-well-specific-barcode]TTVN'-3, Integrated DNA Technologies) was added to each well with Mosquito, and plate was heated at 55°C for five minutes before being immediately placed on ice for two minutes. To each well, 2 μl of first-strand reaction mix (1 μl SuperScript IV buffer, 0.125 μl DTT 10mM, 0.125 μl dNTPs 10mM, 0.125 μl RNase inhibitor (ThermoFisher) and 0.125 μl SuperScript IV reverse transcriptase (ThermoFisher)) was added to each well with Mosquito, and the plate was put through the following reverse transcription thermal schedule: 4°C for 2min, 10°C for 2min, 20°C for 2min, 30°C for 2min, 40°C for 2min, 50°C for 2min and 55°C for 10min. Cells were then pooled manually with a multichannel pipette, mixing thoroughly before uptake to ensure cells were resuspended. To pooled cells, 1.8 μl of 1mg/ml DAPI (ThermoFisher) was added. 50 cells per well were then sorted into four 96-well plates with 3.6 μl water and 0.4 μl NEBNext Ultra™ II Non-Directional RNA Second Strand Synthesis buffer in each well. This was done with a MoFlo XDP, with a 100 μm nozzle, gating on scatter and DAPI to stringently remove doublets and debris. To each well of each plate, 1 μl of second strand synthesis mix using NEBNext Ultra™ II Non-Directional RNA Second Strand Synthesis Module (0.65 μl water + 0.1 μl NEBNext second strand synthesis buffer + 0.25 μl NEBNext second strand synthesis enzyme) was added. Plates were vortexed at approximately 1000rpm for five seconds and spun down for five seconds, and were then incubated at 16°C for three hours. Plates were then stored with foil lids at -80°C for up to a month. Tagmentation and clean up were performed manually with a multichannel pipette, two plates at a time. For tagmentation, plates were thawed and 5 μl tagmentation mix (4.875 μl tagmentation buffer + 0.125 μl Tn5 4nM) was added. Plates were incubated at 55°C for five minutes. Then, 10 μl DNA-binding buffer (Zymo) was added and plates were incubated at room temperature for five minutes to allow for quenching of tagmentation reaction and lysis of cells. Next, 30 μl Ampure SPRI beads (Beckmann Coulter) was added to each well. Plates were vortexed at 1000rpm for 10 seconds and incubated at room temperature for five minutes. Plates were added to magnets, 42 μl supernatant was removed and each well was washed with 100 μl 80% ethanol twice. Plates were removed from magnets and left to air dry for 2–3 minutes. 10 μl EB buffer (Qiagen) was added to each well, and plates were vortexed at 2000rpm for one minute and spun briefly for seven seconds, before being left to incubate at room temperature for ten minutes. Plates were returned to magnets, and 9 μl supernatant was moved to a 384-well Lobind plate. 1 μl PCR primer mix containing 10 μM P5 primer (5'-AATGATACGGCGACCA CCGAGATCTACAC[i5]ACACTCTTTCCCTACACGAGCTCTTCCGATC T-3'; IDT) and 10 μM P7 primer (5'-CAAGCAGAAGACGGCA TACGAGAT[i7]GTCTCGTGGGCTCGG-3'; IDT) was added to each well of 384-well plate with the Mosquito, and 10 μl NEBNext High-Fidelity 2X PCR Master Mix was added manually with a multichannel pipette. PCR was performed with the following program: 2C for 5min, 98C for 30s, and 10 cycles of 98 $^{\circ}\text{C}$ for 10s, 66 $^{\circ}\text{C}$ for 30s, 72 $^{\circ}\text{C}$ for 1min and a final 72 $^{\circ}\text{C}$ for 5min. The full library was then pooled for a total volume of around 7ml. This library was distributed into 8 samples of 850 μl . For each, an Ampure SPRI clean up was performed with 0.9X beads, eluting each sample in 100 μl . These eluted samples were collected into two 400 μl samples. To these samples, double size selection was then performed: 200 μl SPRI beads was added, incubated for ten minutes and then added to a magnet. 600 μl of supernatant was moved to new tubes, where 160 μl fresh beads were added to sample. After 5 minutes of incubation, samples were again added to magnet, and washed twice with 1ml 80% ethanol. The two samples were each eluted in 30 μl for 60 μl total library product. The final library product was quantified using Qubit and BioAnalyzer. For initial pilot of sci-FATE protocol, library was sequenced on NextSeq 500 with 300M reads. All subsequent pilots were sequenced on NovaSeq SP with 700M reads per sample. For experiments, libraries were sequenced two samples at a time on NovaSeq S1, total approximately 900M reads per sample (75,000 raw reads per cell). Sequencing was done with configuration 18-10-10-130 (read1-index1-index2-read2). A full list of primer sequences used is available as a supplementary table.

Tn5 production and transposase assembly

Tn5 was produced as per,^{67,77} to make 10 μ M stock solution. In brief, Tn5-E54K-L372P was expressed and purified using affinity and size-exclusion chromatography. The pETM11-Sumo3-Tn5E54K,L372P plasmid was obtained from EMBL-Heidelberg. Transformation was performed into BL21 (DE3) Gold E. coli (Agilent).

Transposase assembly was performed as per (Martin et al. 2022). In brief: Tn5-N7 (5'-G TCTCGTGGGCTCGGAGATGTGTATAA GAGACAG-3') and Tn5-ME (5'-[phos]CTGTC TCTTATACACATCT-3') oligos were resuspended in annealing buffer (50 mM NaCl, 40 mM Tris-HCl pH8.0) at 100 μ M. 25 μ l of each oligo was mixed, and annealed with the following thermocycle program: 95°C for 5 minutes; cooling to 65°C at 0.1°C/s, then held at 65°C for 5 minutes; then cooled to 4°C at 0.1°C/s. Stock Tn5 was then diluted to 4 μ M, and 20 μ l of this Tn5 was mixed with 20 μ l of the annealed adaptor mix. This Tn5/adaptor mix was incubated at 23°C for 30 minutes, shaking at 300rpm. 20 μ l glycerol was then added and mixed thoroughly, before storing at -20°C. Tagmentation buffer for further dilution was made in a glass bottle on ice: to 38.5ml of nuclease-free water, 1ml 1M Tris pH7.6, 250 μ l 2M magnesium chloride, and 10ml of dimethylformamide were added (with DMF measured out in a glass measuring cylinder). 1.25ml aliquots were made and stored at -20°C.

Species mixture experiment

Mouse HM1 ES cells were cultured as per^{46,47,67} in ES cell medium with 1,000U/ml LIF on feeder cells and human H9 cells in StemFlex (ThermoFisher) on a laminin coating. Both were fixed in methanol + 0.1% DEPC as per above and stored at -80C. Cells were moved to ice for three minutes, then spun down at 2000rpm and 4°C for 5 minutes. Cells were resuspended in 1ml PBS + 0.1% DEPC + 3% v/v BSA, spun down again as before and resuspended in PBS at 1 million cells/ml. Mouse and human cells were then mixed in equal proportions and combinatorial indexing library preparation was performed as above.

Cell loss quantification

Day 4 differentiation cells were collected and fixed in methanol as above. The modified chemical conversion protocol was carried out as described above, the original chemical conversion protocol was carried out as per.²⁵ Note that for this comparison, methanol fixed cells were used for both protocol, while the original protocol used cells fixed in 4% PFA. This was done to prevent differences in fixation from confounding differences in chemical conversion steps — however we and others have observed similar low yield of cells with original protocol and PFA-fixed cells.²⁵ For both, cells were initially counted and diluted to 5 million cells before chemical conversion steps, and the number of cells remaining was counted after all steps completed.

sci-FATE pilot

The initial pilot of sci-FATE was performed with in vitro differentiating ES cells (day 4 of protocol), exactly as described in²⁵, with the following changes: i) no 4sU labelling was performed, ii) we ran 19 PCR cycles, iii) to concentrate the library, we collected 8 x 800 μ l aliquots of library, and performed two rounds of AMPure bead concentration (0.7X then 0.8X) for a final library volume of 100 μ l.

RNA-seq and RT-qPCR

For both bulk RNA-seq and RT-qPCR, samples were collected and RNA isolated with a Qiagen RNeasy kit. For RNA-seq, samples were normalised to 375ng and library prep was performed with Kapa mRNA Hyperprep with polyA capture beads (KK8421). Libraries were sequenced using NovaSeq6000 with 25 million reads per sample. Differential expression analysis was performed with DESeq2,⁷³ with three replicates per condition. For untreated control, separated sets of untreated samples were compared. For RT-qPCR, 1 μ g of RNA was reverse transcribed using Super-Script III first strand synthesis kit (ThermoFisher) with random hexamers. PowerUp SYBR Green Master Mix (ThermoFisher) was used to perform qPCR with tenfold diluted cDNA, using actin as a control.

Martin et al. protocol

For sci-RNA-seq2 comparisons, in vitro differentiating ES cells (day 4 of protocol) were fixed in methanol as described above. For library preparation, cells were washed once in 1ml PBS + 0.1% DEPC (where?) + 3% v/v BSA (NEB) then diluted to 1 million cells per ml in PBS and immediately submitted for library preparation. We performed combinatorial indexing protocol as above. For Martin et al. 2022 protocol,⁷⁶ after second-strand synthesis, we added 1 μ l Qiagen protease per well and incubated for 30 minutes at 37C, then 20 minutes at 75C. We then performed tagmentation as above, before adding a quenching mix of 0.375 μ l BSA (NEB), 0.375 μ l 1% SDS and 2.25 μ l water to each well, incubating at 55C for 15 minutes. We then added 4 μ l 5% Tween-20 to each well, proceeding to PCR with the addition of 2 μ l primers and 20 μ l NEBNext PCR mix. For the sample shown, we ran 16 PCR cycles, collected 3 μ l per well, performed a 0.8X volume AMPure bead clean and isolated 200–600bp DNA by running the library through a 1% agarose gel and gel extraction (Qiagen), as performed in Martin et al. 2022. In parallel, we ran an additional sample which we purified through a round of 0.8X AMPure bead cleaning followed by purification through a size-selection DNA column (Zymo) – we did not observe major differences in the data quality through this second approach.

Data processing and label detection

Raw data processing and label detection was performed with the dynast package²⁷ and the GRCm39 reference genome. To estimate SNPs (and background conversion rate for estimation), dynast was run in 'control' mode, with a sci-RNA-seq2 sample from all time points of the differentiation mixed together, collected without 4sU labelling or chemical conversion. We found little benefit to using dynast's consensus calling feature, so did not use it for analysis.

Previous work has found that 4sU incorporation rate is roughly 1 in 40 uridines,²⁸ implying that a significant proportion of nascent reads would not possess a detectable 4sU label within the sequenced read. This has previously been resolved with a Bayesian approach^{27,78} to estimate the experimental T-to-C conversion rate and infer the new-total ratio for each gene in each cell with a binomial mixture model. This Bayesian approach is implemented in *dynast* with two modes: ‘ π_g mode’, which implements the original approach to infer a parameter for each gene in each cell, or α mode, which instead infers a cell-wise correction value by which the number of labelled reads is adjusted across all genes for a cell. The motivation for the α mode is to make the problem more tractable for the sparse, noisy data of single-cell sequencing. We found that *dynast*’s π_g implementation failed even basic tests (such as displaying a positive correlation between new and total reads), and so we progressed with α mode instead. With this method, we observed an increase in the proportion of reads labelled as new from 20% to 40%, consistent with the expectation that as many as half of new reads would not contain a detectable label (Figure S8D). However, visualising new vs. old reads for estimation and counting methods for a given gene (Olig2, for example, in Figures S8B and S8C) showed a large increase in the number of cells predicted to possess only ‘new’ reads, and a reduction in the proportion of cells that possess both new and old reads (and thus, contain useful dynamical information) (Figures S8A and S8B). We also found that across all genes, with estimation there was a consistent positive correlation between the total number of reads detected in a cell and the label rate detected in a cell, which was absent when using the counting method (Figure S8E). We found that steady-state degradation rate approximations from counting data gave a half-life distribution closer to the experimentally measured half life distribution of mRNAs in mES cells,⁷¹ when compared to degradation rates inferred from estimated labelling (Figure S8F). Finally, testing the dynamical information with estimation vs. counting with VelvetVAE and *dynamo*, we found that using estimated label rates substantially worsened velocity predictions, as assessed by visual inspection (Figures S8G–S8I).

Based on these findings, we chose not to use the Bayesian estimation method for our analysis, finding that inference of dynamics generally performs well without it. Our velocity model is defined in terms of observed new and total reads, along with a degradation parameter that is itself initially estimated from the observed new reads. As such, velocity is defined relative to the observed new, and the effect may only be that the magnitude of velocity vectors may be underestimated.

Moreover, our results are not entirely consistent with the observation that as many as half of nascent reads may not be detected as labelled. It may be that recorded 4sU incorporation rates are underestimations: these figures come from studies that look at incorporation rates in bulk RNA²² (which would include many species such as rRNA with considerably longer half lives than mRNA) or with below-saturation levels of labelling²⁴ (e.g. 100 μ M) for labelling periods of only 1–3 median mRNA half lives (12–24 hours). As a result, 4sU may not have fully saturated the transcriptome in these studies, leading to an underestimation of incorporation rate. More work on the measurement and Bayesian inference of 4sU incorporation is needed.

Preprocessing

Basic processing was performed with ScanPy.⁷² To remove low quality cells, we set lower and upper thresholds at the tenth and ninetyth percentiles for number of genes and UMIs observed per cell, and removed data that fell below or above these thresholds, respectively. This corresponded to an acceptable range of approximately 4000–25000 UMIs/cell and 2000–7000 genes per cell. We removed cells with less than tenth percentile labelling rate (approx. 10% labeling) and more than ninetyth percentile mitochondrial percentage of reads (approx. 5%). We used scrublet⁷⁹ to remove cells with a doublet score of above 0.3. To handle batch effects, highly variable gene selection was stratified by replicate: 3000 highly variable genes were selected per replicate (excluding replicate 4 as an incomplete replicate), and only genes selected across all three replicates were retained (generally around 2000 genes). We found this successfully removed an evident batch effect between replicate 1 and replicates 2 and 3 (Figure S2H). For cell type classification, we adopted an ensemble approach: we clustered cells with the Leiden algorithm with high resolution (1.6), producing many more clusters than expected cell types (32 clusters, 9 expected cell types). We performed differential expression (using ScanPy) between clusters, scored cells based on key marker genes of expected cell types, and performed a basic classification based on binarised expression comparison to a knowledge table of marker genes, as per.^{2,14} For each cluster, we visualised the cluster in UMAP and MDE embeddings, visualised the key-gene scores, the marker-based classification distributions and the top differentially expressed genes, and based on all analyses together, clusters were assigned to one of 10 groups: NMP, mesoderm, early neural, neural, motor neuron progenitor (pMN), MN, p3, V3, floor plate or other.

Analysis of RNA velocity

For analysis of RNA velocity, we used *scVelo* to access RNA velocity datasets from human bone marrow, mouse dentate gyrus, mouse pancreas, and additionally accessed RNA velocity data from human and mouse neural tube.^{2,13–15,48,49} For visualisation of velocity data, we ran *scVelo* on default parameters, projecting velocities either using *scVelo*’s projection method, or by calculating directly projected velocity as:

$$V_{proj_X} = (X+V)_{proj_X} - X_{proj_X}$$

Where $proj_X$ represents the application of $\log_1 P$ transformation followed by projection to PCA embedding of X .

Comparison and benchmarking

To visualise VelvetVAE inference, data was split into two subsets: neural (early neural, neural, pMN, p3, floor plate, MN and V3) and NMP (NMP, mesoderm, early neural, neural and pMN from D3.2 to D5). VelvetVAE was trained on default settings for both. As comparison, dynamo and scVelo were run on labelling and splicing data of the same systems respectively with default settings. For all three, visualisation was performed by predicting high dimensional velocities and directly projecting these velocities to PCA space.

For quantitative benchmarking, we split the dataset into six subsets: i) pMN/MN, (approx. 8,000 cells) ii) p3/V3 (7,000 cells) iii) mesoderm (10,000 cells) iv) NMP system (18,000 cells) v) neural system (20,000 cells) vi) whole dataset (45,000 cells). These subsets were chosen to capture different resolutions of dynamics, from linear trajectories to multifurcations. For each subset, scFates⁵⁶ was used to construct a pseudotime skeleton based on known biological trajectories. The parameters used for scFates were manually tweaked to ensure faithful recapitulation of known biological dynamics. With this skeleton, two methods were used to create a ground truth: i) PTS: each cell was mapped to the closest position on the skeleton, and velocity was set as the displacement between that position and the subsequent position, ii) CRS: a pseudotime value for every cell was determined using scFate's pseudotime function, and this was passed to CellRank's PseudotimeKernel,⁵⁷ which was used to calculate a transition matrix which was then projected to PCA. For subset (vi) (whole dataset), an accurate pseudotime could be computed, but the underlying trajectory skeleton was not deemed to be accurate enough, so this dataset was not used for PTS assessment. Both ground truth velocities existed in 50D PCA space, and all tested velocities were also projected to 50D PCA space for comparison through cosine similarity. This projection was done directly, not using a nearest-neighbour projection heuristic. The reason for evaluating velocities in 50D PCA space, rather than high dimensional gene expression space, is because as the number of dimensions increases, the probability that any two random vectors in that space happen to be almost orthogonal also increases. In other words, cosine similarity measures suffer from the 'curse of dimensionality,' and cosine similarities in thousands of dimensions can be noisy and hard to interpret. A third metric was used, cross-boundary direction (CBD), as done previously, using the implementation from.³³ With this method, data was clustered with Leiden algorithm, and directions between specific clusters was defined. The agreement between inferred velocities and these directions was assessed by cosine similarity.

Gene-specific scores were calculated as follows: Velocities in the relevant data subset were calculated, and for the defined gene in the defined cell type, the percentage of velocities that had the correct direction (positive or negative) was calculated. In total, twenty different gene-specific velocity comparisons were made, across five cell types. The full list of these gene-specific comparisons is available as a supplementary table.

For other velocity tools, velocity inference was run on default settings where possible. Where default settings failed to train or returned errors, settings were adjusted based on available tutorials and documentation. Additional to the velocity inference tools tested in this study, we attempted to include scDVF/DeepVelo³⁶ and CellDancer⁸⁰ but were unable to achieve working velocity inference with either tool.

VelvetVAE and SvelvetVAE

VelvetVAE can be thought of a mechanistic variational auto-encoder that simultaneously learns to map total expression to a latent representation and to predict latent velocities that can construct observed labelled expression. In this sense, the VAE component is a generative model, and the velocity inference is a form of supervised prediction (predicting labelled data via latent dynamics and a biophysical equation). Only total expression is encoded in the VAE, not labelling data.

The framework for VelvetVAE inference from metabolic labelling starts from the two equations:

$$\frac{dN}{dt} = \alpha - \gamma N \quad (\text{Equation 1})$$

$$\frac{dX}{dt} = \alpha - \gamma X \quad (\text{Equation 2})$$

These represent the time derivatives for labelled (N for 'new') and total (X) reads, respectively, based on transcription (α) and degradation rates (γ). This framework assumes that the transcription rate for labelled and unlabelled reads is the same, for more details see supplementary note 1. Since we can assume no initial conditions (i.e. the amount of labelled reads when we introduce the label at $t = 0$ is zero), we can solve the labelled derivative equation to give:

$$N = \frac{\alpha}{\gamma} (1 - e^{-\gamma t}) \quad (\text{Equation 3})$$

This can be rearranged to give an equation for transcription rate:

$$\alpha = \frac{N\gamma}{1 - e^{-\gamma t}} \quad (\text{Equation 4})$$

Inserting this into the equation for total velocity, we get our key equation:

$$\frac{dX}{dt} = \frac{N\gamma}{(1 - e^{-\gamma t})} - \gamma X \quad (\text{Equation 5})$$

This defines velocity in terms of three observables (new reads, N ; total reads, X ; labelling time, t) and one latent parameter, the degradation rate, γ . Unlike RNA velocity, there is no need to infer splicing parameters, initial conditions or gene-specific latent times. This equation can be rearranged to provide a prediction of labelled/new reads based on predicted total expression and velocity:

$$N = \frac{1 - e^{-\gamma t}}{\gamma} \left(\frac{dX}{dt} + \gamma X \right) \quad (\text{Equation 6})$$

This equation is built into the loss function of VelvetVAE. For SvelvetVAE, we start with a similar framework:

$$\frac{dU}{dt} = \alpha - \beta U \quad (\text{Equation 7})$$

$$\frac{dX}{dt} = \alpha - \gamma X \quad (\text{Equation 8})$$

We assume $\beta = 1$ as per previous work.¹³ At steady-state (when $\frac{dU}{dt} = \frac{dX}{dt} = 0$), these equations can be equated to give $\gamma = \frac{U_{ss}}{X_{ss}}$, providing a velocity definition:

$$\frac{dX}{dt} = U - \gamma X \quad (\text{Equation 9})$$

Which, as with VelvetVAE, is reframed to predict unspliced reads from predicted total expression and velocity. This is a minor distinction compared to previous velocity approaches which model unspliced-spliced ratio (rather than unspliced-total proportion). This is done to bound γ between zero and one, with the aim of improving stability in the face of highly variable unspliced proportions across genes (Figure 1A).

The VelvetVAE framework is built as a variational autoencoder (VAE). The core structure of this model builds on implementations of scvi-tools.^{54,55} Briefly, the VAE is composed of two neural networks, an encoder and a decoder. The encoder maps the input data to a lower-dimensional ‘latent’ space. This mapping is probabilistic, meaning that the encoder learns to map to mean and variance parameters of latent variables, assumed to be independent Gaussians, and a latent representation of the data is sampled from these learned Gaussian distributions. The decoder reconstructs the original data from this sampled latent representation, by mapping from this latent space to parameters of a zero-inflated negative binomial distribution. The training involves optimizing the parameters of the encoder and decoder networks through a variant of stochastic gradient descent, where the objective is to minimize the reconstruction loss (the difference between the input and the reconstructed data) and a regularization term, the Kullback-Leibler (KL) divergence, that forces the learned distribution to be close to a standard normal distribution. This dual-objective function effectively results in a balance between data fidelity and statistical regularization. In training, VelvetVAE maps total expression X to a probabilistic latent space Z through an encoder neural network, ϕ , and passes samples from this space through a decoder network, ψ , to produce a reconstruction of the data $\hat{X}$. In parallel, Z is fed into a neural vector field, f_θ that predicts latent velocities, V_Z , which are also passed through the decoder to produce a velocity reconstruction (through the equation $\hat{V} = \psi(Z + V_Z) - \psi(Z)$). Expression and velocity reconstructions are used to predict labelled data based on Equation 6. In this stage of training, there are thus two components to the loss function. First, the conventional loss for a VAE:

$$\mathcal{L}_1(x) = -E_{q(z|x)}[\log p(x|z)] + KL(q(z|x)||p(z)) \quad (\text{Equation 10})$$

Second, the velocity loss:

$$\mathcal{L}_2(n, \hat{n}) = \|\log n - \log \hat{n}\|^2 \quad (\text{Equation 11})$$

By default, the full loss function is $\mathcal{L} = \mathcal{L}_1 + 10\mathcal{L}_2$.

To train either VelvetVAE or SvelvetVAE, we inputted new and total datasets, along with precomputed neighborhood indices for the neighbourhood constraint. By default, VelvetVAE is trained in two stages. First, the VAE and the vector field are trained together, then the VAE parameters are frozen and VAE loss removed, and the vector field is trained with the neighborhood constraint (see below). By default, the first stage is performed for 200 epochs and the second for a further 800. We use the entire dataset as one batch, and use a learning rate of 0.001, and the AdamW optimiser with weight decay of 0.001. By default, we use 50 latent dimensions, a linear decoder, and perform the neighborhood constraint in latent space.

To initially approximate γ with labelling data, we start with the assumption that steady-state dynamics are observed, where $N = kX_{ss}$ and $X_{ss} = \frac{\alpha}{\gamma}$, where k is a proportional value representing the proportion of total that is labelled at steady state. Plugging these into Equation 4 and rearranging for γ gives us:

$$\gamma = \frac{-\ln(1 - k)}{t} \quad (\text{Equation 12})$$

Thus, we perform extreme-value regression of new and total reads as per previous studies,^{31,53} and calculate γ from the estimated slope from these regressions, k .

For splicing data, we cannot define an equivalent of Equation 4 without initial conditions or splicing parameters, so we use a more simplistic approximation of γ , performing unspliced-total extreme-value regression, and setting $\gamma = k$.

Neighbourhood projection

For both the VelvetVAE's neighbourhood constraint and VelvetSDE's Markov random walk, a cell's velocity vector is projected onto displacement edges of the cell and neighbouring cells to provide a 'transition probability' vector. This is done in the same manner as previous studies.^{27,53,57} In brief, we calculate these transition probabilities as follows: we calculate a transition matrix, π , from each cell i to its neighbours:

$$\pi_{ij} = \frac{\exp\left(\frac{\cos(z_j - z_i, z_{vi})}{\sigma^2}\right)}{\sum_l \exp\left(\frac{\cos(z_j - z_i, z_{vl})}{\sigma^2}\right)} \quad (\text{Equation 13})$$

This is the cosine similarity between a cell's velocity vector and the displacement between the cell and its j th neighbours, subject to a softmax transformation that converts these scores to transition probabilities. For Markov random walks, this transition probability is used to govern walks. For neighbourhood projection, we calculate the velocity vector's expected displacement with respect to the cell's transition matrix, as a sum of neighbour displacements weighted by transition probabilities and adjusted for non-uniform data density as per.^{27,53}

Neighbourhood constraint

In the second stage of VelvetVAE training, we combine latent velocity prediction with a neighbourhood constraint. We compare the neighbourhood-based expected displacement with the original velocity vector with a cosine embedding loss:

$$\mathcal{L}_3(V_z, K) = 1 - \text{CS}(V_z, NP(V_z, K)) \quad (\text{Equation 14})$$

Where CS represents cosine similarity, NP represents the neighbourhood projection function and K the set of precomputed neighbour indices. By default, we use 100 neighbours for this projection - the large number chosen to ensure a sufficient sample of the local data manifold is retained. In this stage of training, the total loss function is thus $\mathcal{L} = \mathcal{L}_2 + \mathcal{L}_3$. Predicted velocity and expected displacement from neighbourhood projection should only differ when the projection fails to accurately reconstruct the original vector, which will occur when predicted velocity vector points away from neighbours. In which case, the neighbourhood constraint will encourage predictions to point towards local neighbourhoods. Cell neighbour indices are pre-computed before training, and neighbourhood projection is performed in a broadcasted fashion, allowing a computationally efficient implementation of the neighbourhood constraint.

VelvetSDE dynamical model

To train VelvetSDE, we embed the trained VelvetVAE vector field as the drift component of a neural stochastic differential equation system^{44,45} of the following form:

$$\frac{dX}{dt} = \mu(X, t)dt + \sigma(X, t)^\circ dW_t \quad (\text{Equation 15})$$

Where $\mu(X, t)$ is a neural vector field, $\sigma(X, t)$ is a noise function and W_t represents a Weiner process of Brownian motion. To reduce memory usage, adjoint-mode gradient computation is used. The SDE is constructed as a Stratonovich integral solved using the midpoint method. This is done as Stratonovich integrals have lower computational cost for adjoint methods than Ito integrals.⁴⁴ For this study, $\sigma(X, t) = k$, where k is a user-defined scalar.

In parallel, we define a Markov process model from the learnt vector field: we calculate a transition matrix, π , as described above. We can define a Markov process random walk as:

$$W_{i+1} \leftarrow \text{Multinomial}(1, \pi_{W_i}) \quad (\text{Equation 16})$$

In words: The next state is chosen as a sample from a multinomial distribution with a probability vector equal to the current state's transition probabilities, defined by π . We thus iteratively define random paths through the data manifold determined by the velocity-guided transition matrix. We can optionally define terminal states (where $W_{i+1} = W_i$) and can produce a cubic spline of the Markov path so that the number of timesteps can equal that of nSDE simulations, without requiring that the same number of Markov steps and nSDE steps are taken.

To train VelvetSDE, for each cell in a batch, we simulate n trajectories with the nSDE model (X^{SDE}) and n trajectories with the Markov process random walk (X^{MRW}). Each cell's loss is defined as the average of the Kullback-Leibler divergence of the SDE simulation from the Markov random walk, stratified by timestep:

$$L(X^{SDE}, X^{MRW}) = \sum_i D_{KL}(X_{t=i}^{SDE} || X_{t=i}^{MRW}) \quad (\text{Equation 17})$$

In words, for each timestep, this is the Kullback-Leibler divergence of that timestep's distribution of nSDE points from that timestep's distribution of Markov simulations, assuming both distributions to follow a Gaussian distribution. This measures how far from

the Markov simulation the nSDE simulation has moved. For training, we set the noise magnitude of the nSDE to roughly equate the noise magnitude of the Markov random walk (assessed through visual inspection of trajectories in PCA space). In this study, a noise magnitude of 0.2 is used for this. We train for 250 epochs, running 200 cells per epoch and 50 simulations per cell (each, for nSDE and Markov random walk), with same learning rate and optimiser as above.

Decision boundary analysis

To cluster trajectories, we performed k-means clustering on trajectories, reshaped to be $(n, t \times d)$ where n is the number of trajectories, t is the number of timesteps and d is the number of latent dimensions. To define decision boundaries, we set noise magnitude to 0.1, simulated each cell's trajectory ten times (chosen arbitrarily; changing this number affects the size of mixed-fate population, as can changing the model's noise magnitude), clustered all trajectories, and define 'mixed-fate' cells within a decision boundary as any cell with trajectories belonging to multiple clusters. To investigate the role of noise in the temporal evolution of decision boundaries, we performed neighbourhood smoothing of timepoints to create a continuous scale of time, binned cells into time windows, and performed the above decision boundary analysis for each time window across multiple levels of noise magnitude in VelvetSDE's diffusion component. Time smoothing was performed to accommodate the uneven distribution of timepoints (D3.2, D3.4, D3.6, D3.8, D4 and D5), and analysis provides a proof of principle, though the specific timings may be biased by this uneven sampling. For *in silico* perturbation, we took 'mixed-fate' cells, and set each TF's expression value in turn to the maximal observed value for that TF in mixed-fate cells, we then projected these perturbed data to the VelvetSDE latent space, and simulated each cell's trajectory. We classified trajectories by assigning each trajectory to the pre-defined trajectory cluster (i.e. neural or mesoderm) that it is on average closest to. To determine negative log p-values, we performed binomial tests, where k =observed number of neural trajectories, n =number of simulated trajectories and p =observed unmodified probability of neural trajectories from mixed-fate cells. Note that we tested setting gene expression to zero, rather than maximums, and found the expected inverse results (as expected) however trends were less clear, possibly due to the confounding factor of drop-out being modelled in VelvetVAE's zero-inflated negative binomial fitting (in other words, zero expression is caused by lack of expression and by dropout; maximal expression is only caused by maximal expression, thus exploring decision boundaries through maximal expression made more sense).

Gene expression analysis

To produce gene expression distributions from trajectories, we project them through the model's decoder to gene expression space, taking the decoded mean value as the expression level. All trajectories were simulated for 100 timesteps. For Markov random walks, we simulated 50 steps with a transition matrix constructed using 10 nearest neighbours. We used cubic splines to interpolate 50 steps into 100 timepoints. For pseudotime binning, we calculated velocity pseudotime using scVelo,⁵³ and sorted cells into 100 bins based on pseudotime score. For each simulation, cells were randomly chosen from each bin and trajectories constructed as the path across pseudotime bins. To score trajectories on conservation of expected gene orderings, we used seven orderings: *Sox2* → *Olig2*, *Irx3* → *Olig2*, *Pax6* → *Olig2*, *Olig2* → *Neurog2*, *Neurog2* → *Mnx1*, *Neurod4* → *Isl1*, and *Mnx1* → *Tubb3*. To assess ordering, we compared the argmax of each gene being compared for a given trajectory. If a trajectory contained all orderings correct, it scored 1, otherwise 0, and the score reported is the mean across all trajectories for a given method.

Neural patterning analysis

Median trajectories were averaged by taking the median of each timestep across clustered trajectory distributions. Confidence intervals were calculated by grouping trajectories based on the replicate of the initial cell, and averaging before calculating the standard deviation across replicates and using this to calculate 95% confidence intervals.

Computational resources

All analysis was run on a single GPU core from NVIDIA V100 GPU node and 8 CPU cores, with 120Gb of memory. In this configuration, for approx. 20,000 cells, VelvetVAE trains within five minutes, VelvetSDE within one minute.

QUANTIFICATION AND STATISTICAL ANALYSIS

For Figure 4G, to determine negative log p-values, we performed binomial tests, where k =observed number of neural trajectories, n =number of simulated trajectories and p =observed unmodified probability of neural trajectories from mixed-fate cells. For Figure 5E, confidence intervals were calculated by grouping trajectories based on the replicate of the initial cell and averaging before calculating the standard deviation across replicates and using this to calculate 95% confidence intervals.

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Supplemental information

**Reconstructing developmental trajectories
using latent dynamical systems
and time-resolved transcriptomics**

Rory J. Maizels, Daniel M. Snell, and James Briscoe

Reconstructing developmental trajectories using latent dynamical systems and time-resolved transcriptomics

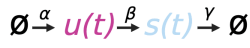
Rory J. Maizels^{1,2}, Daniel M. Snell¹, and James Briscoe¹

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SUPPLEMENTARY MATERIALS

A splicing velocity



$$\frac{du}{dt} = \alpha - \beta u(t) \quad \frac{ds}{dt} = \beta u(t) - \gamma s(t)$$

unspliced
spliced
velocity

labelling velocity

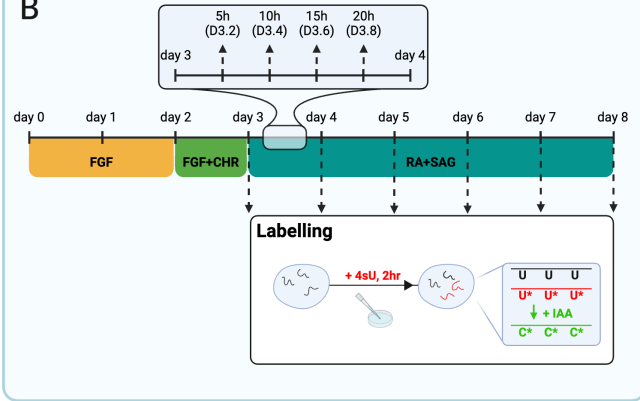


$$\frac{dx}{dt} = \alpha - \gamma x(t) \quad \frac{dn}{dt} = \alpha - \gamma n(t)$$

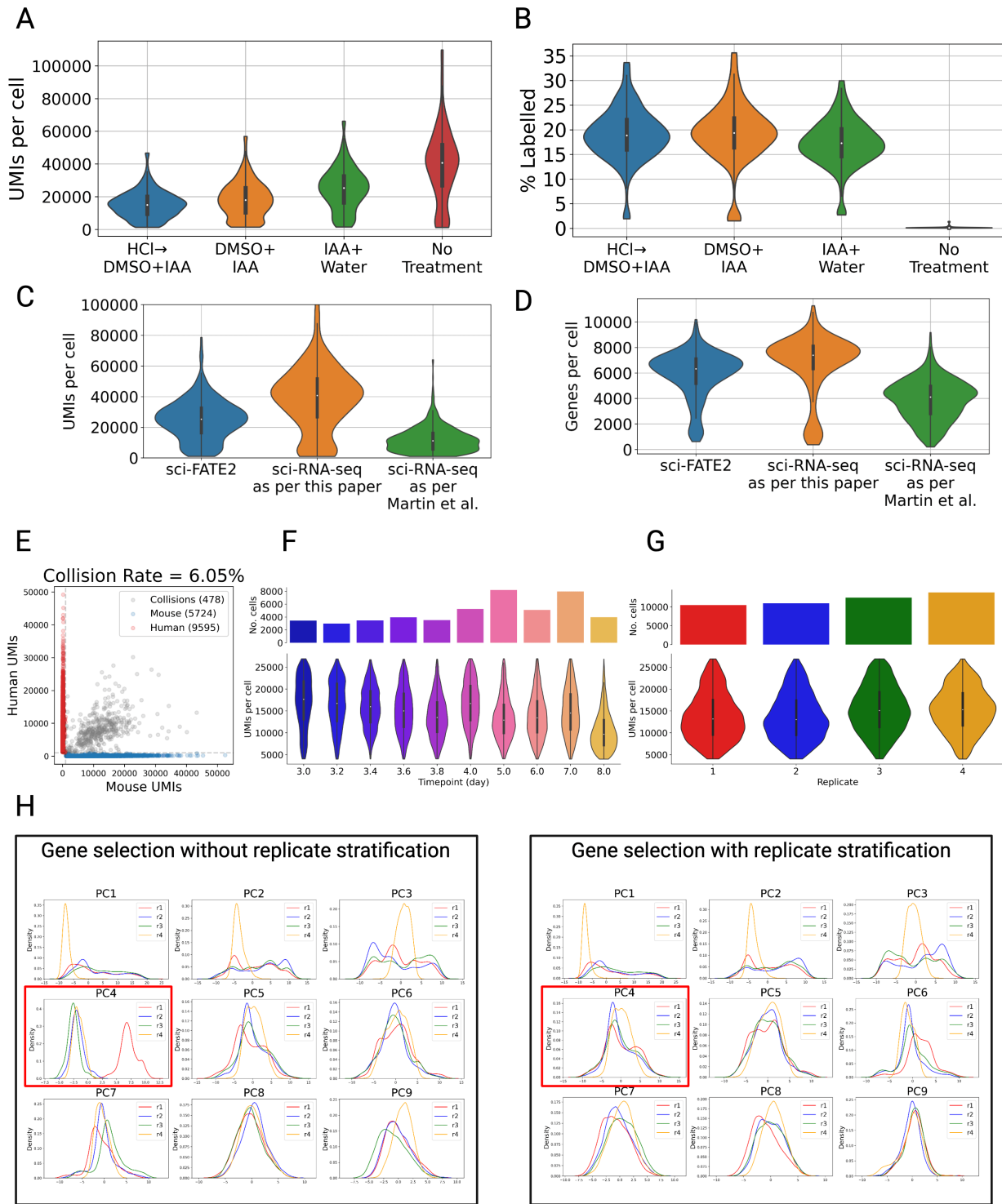
$$\alpha = \frac{n\gamma}{1 - e^{-\gamma\tau}}$$

labelled
total
velocity
labelling time

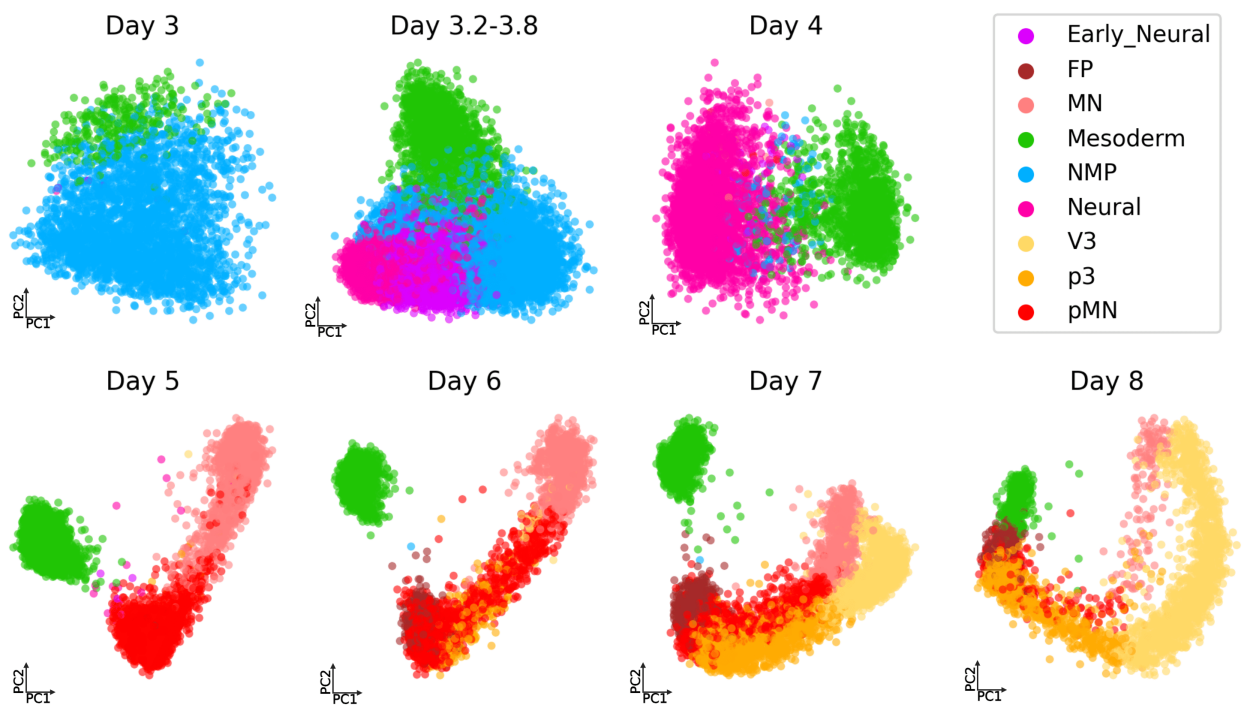
B



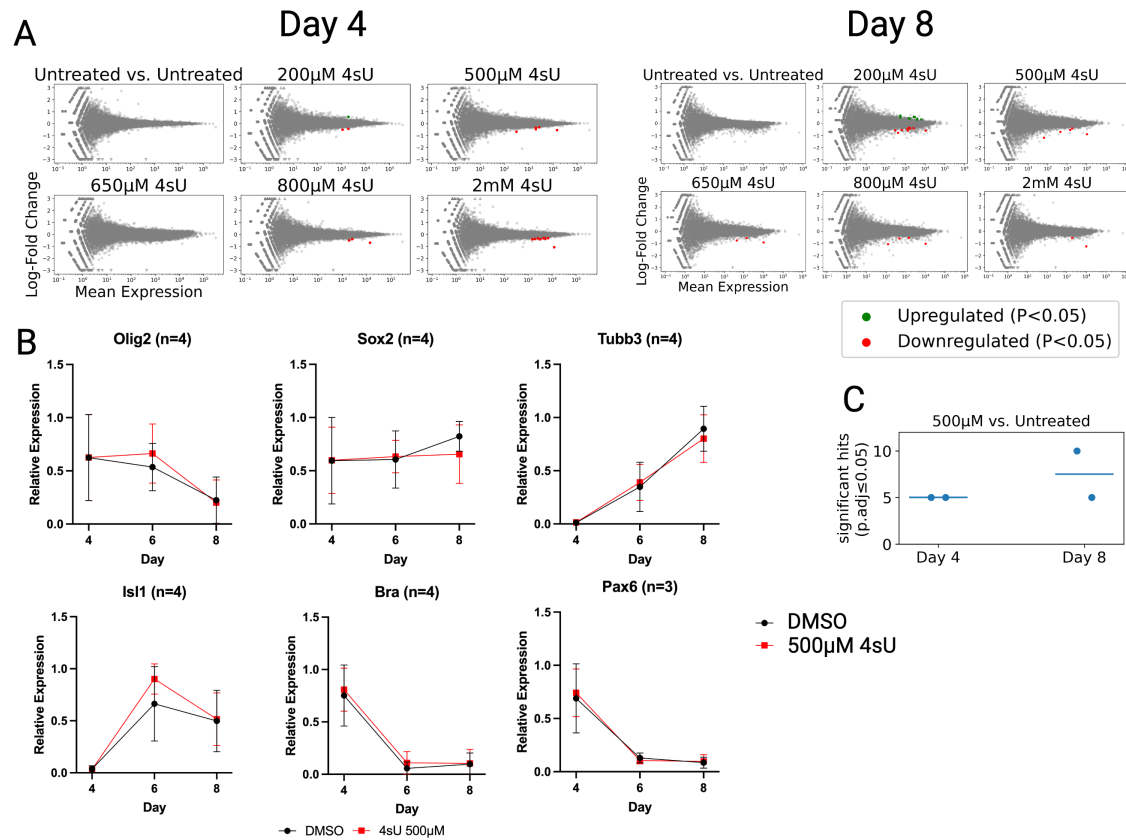
Supplementary Figure 1: Experimental schematics additional to Figures 1 and 2. A. Schematic comparison of velocity frameworks from splicing and labelling data. Labelling does not require the additional consideration of splicing parameters, and has an experimentally determined labelling time instead of an unknown latent time. B. Schematic of experimental set up for this study. In vitro differentiation of ventral neural populations was performed, with collections on days 3 to 8, and every 5 hours between day 3 and 4. For each collection, cells were labelled with 4sU for two hours before collection and fixation.



Supplementary Figure 2: Quality control metrics for sci-FATE2, supplementary to Figure 1. A. UMIs per cell with different IAA treatment protocols. B. Percentage of reads detected as labelled with different IAA treatment protocols. C. UMIs per cell comparison for sci-FATE2 (this study's data), sci-RNA-seq (pilots from this study run without metabolic labelling or chemical conversion) compared to a pilot of a recently published updated combinatorial indexing protocol [1]. D. The same comparison for genes detected per cell. E. Collision rate measured from human-mouse mixture experiment. Collision rate is equal to double the observed data points with both human and mouse reads. F. Number of cells and UMI per cell distribution of this study's data, stratified by timepoint. G. The same data, stratified instead by replicate. H. Demonstration of the use of replicate stratification to remove batch effects without data transformation. When selecting highly variable genes, only genes that were selected as highly variable in each individual replicate were retained. Red box highlights the distributions for different replicates in the fourth principal component of PCA. Without replicate stratification, a clear batch effect is observed; with it, this is removed. Note the difference in distribution between replicate 4 and others is expected, due to this replicate interpolating between the first two timepoints of other replicates.

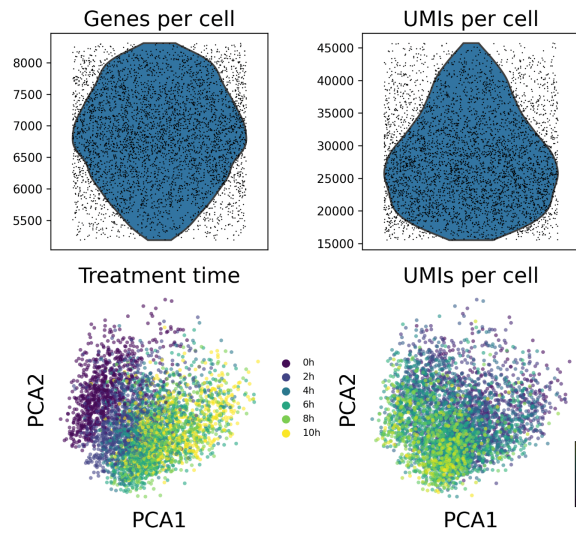


Supplementary Figure 3: Visualisation of each timepoint, supplementary to Figure 1 Plots of the first two principal components for each timepoint in the dataset, coloured by cell type classification.

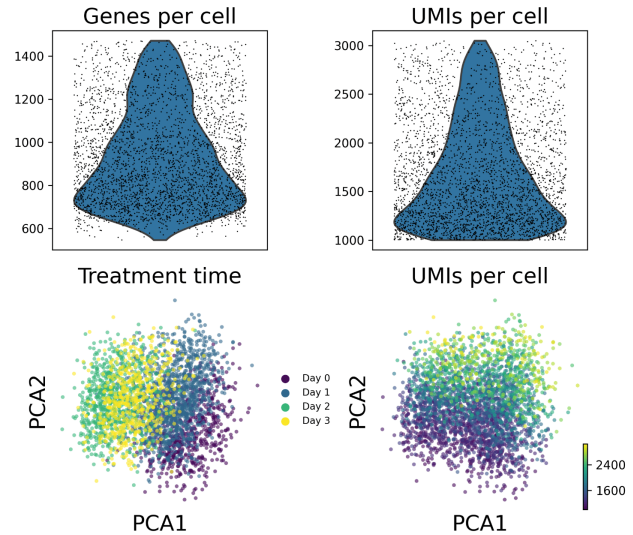


Supplementary Figure 4: Effects of 4sU labelling, supplementary to Figure 1. A. Representative MA plots of RNA-seq at different concentrations of 4sU (all for two hours before collection). Each sample performed in duplicate, with two different untreated populations used for untreated vs. untreated comparison. RNA-seq of samples with and without 2 hours of varying concentrations of 4sU labelling before collection on day 4 and 8 of differentiation. Green points are genes significantly increased in 4sU conditions, red are significantly decreased. B. RT-qPCR of key motor neuron genes with and without 2 hours of 500 μ M 4sU labelling directly before collection. For relative expression, each replicate was normalised between 0 and 1. C. Summary of RNA-seq results for 500 μ M 4sU for two hours, across two replicates.

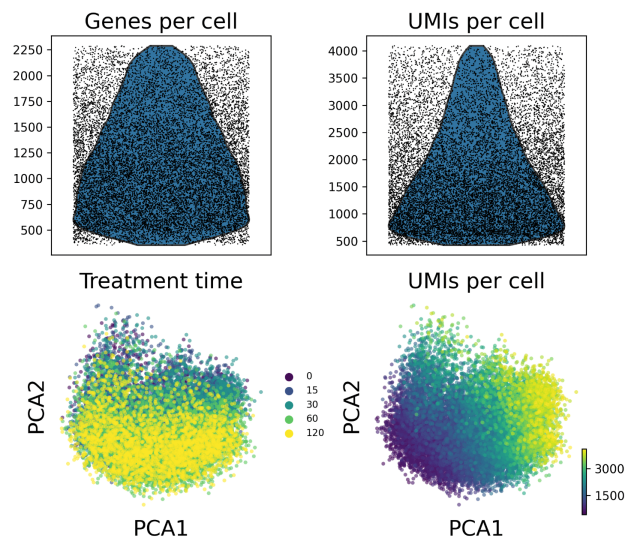
A sciFATE QC (after filtering)



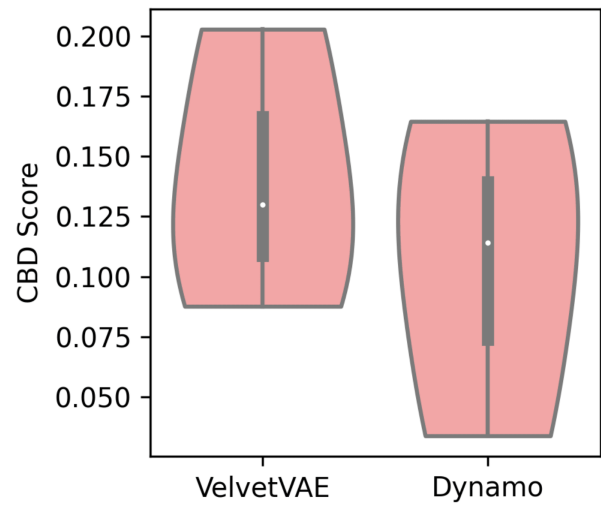
B Well-temp-seq QC



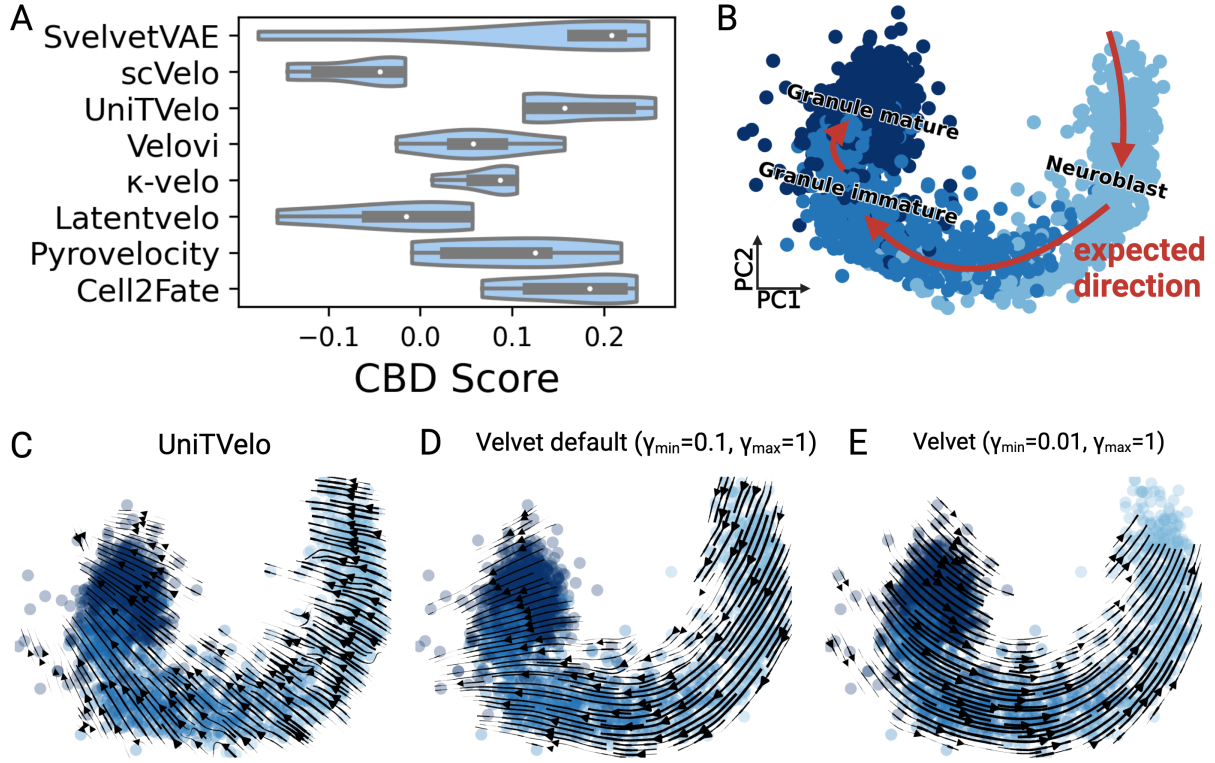
C scNT-seq QC



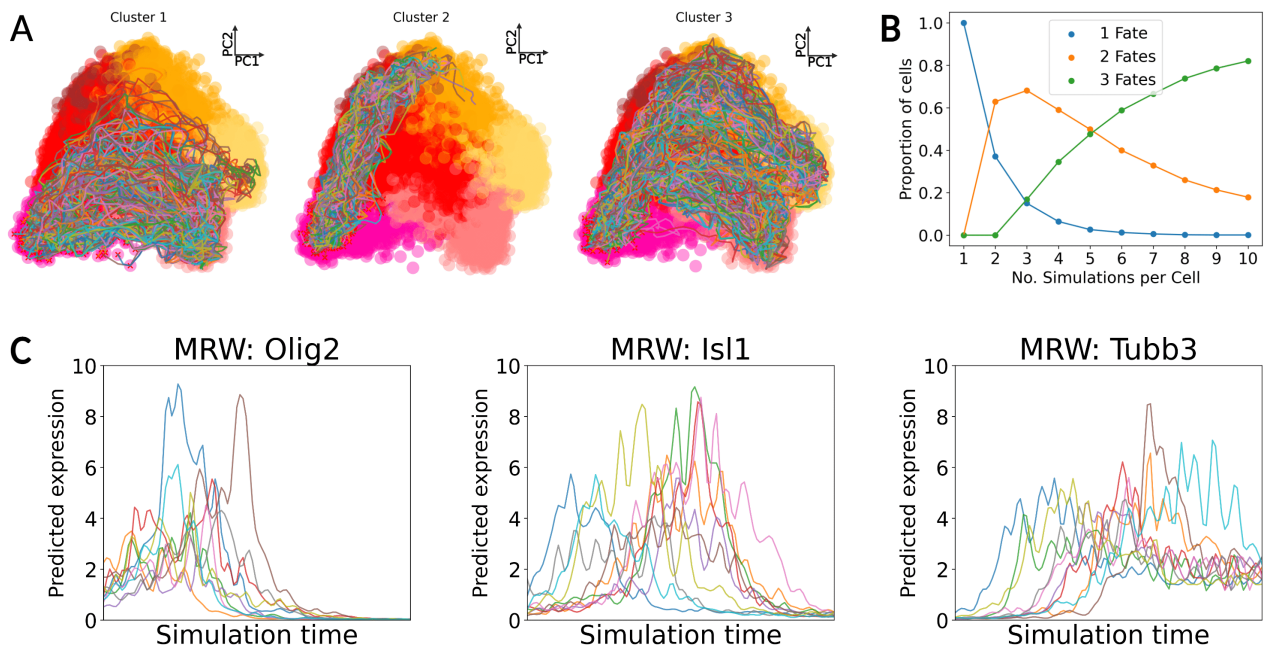
D



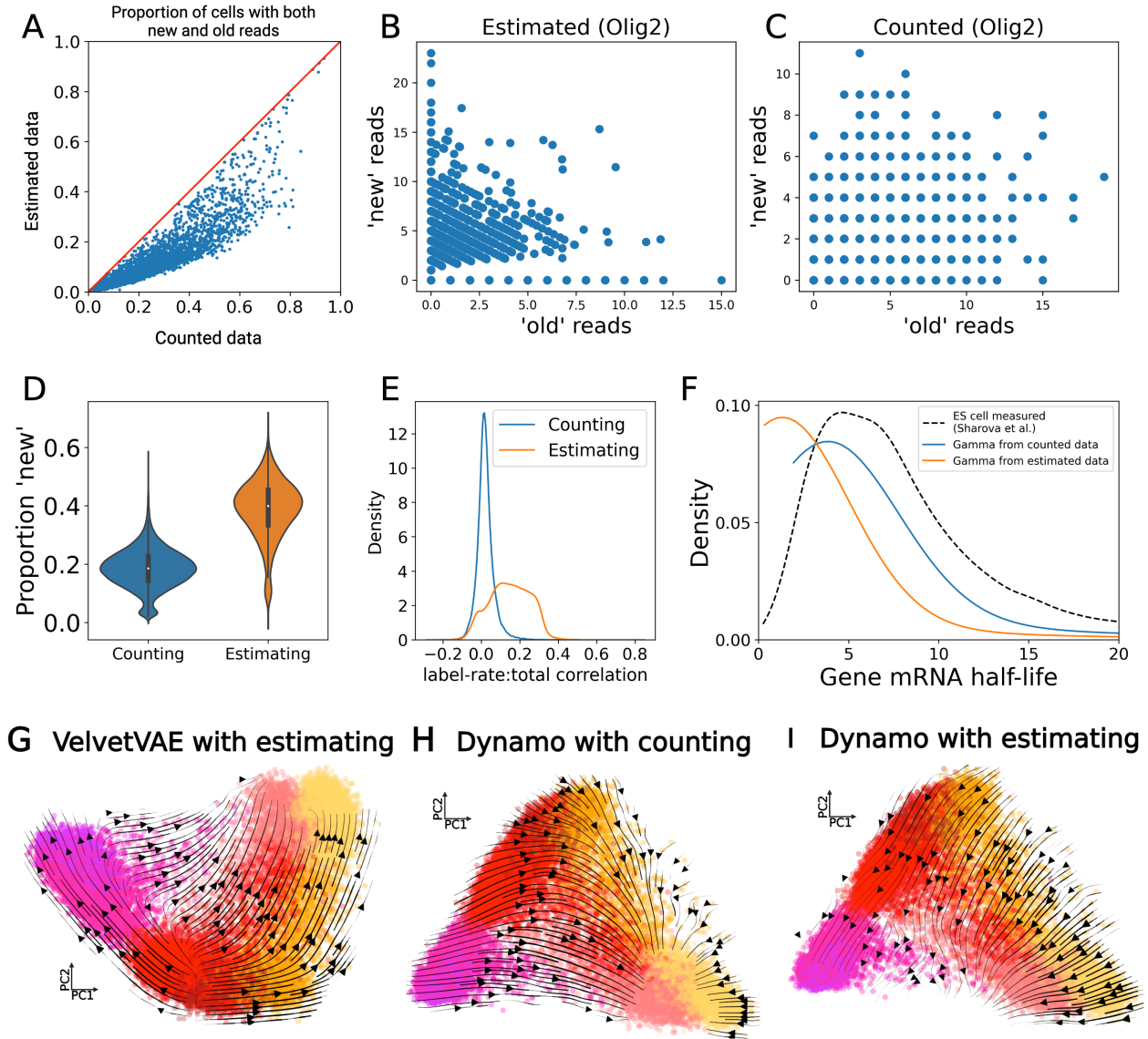
Supplementary Figure 5: Benchmarking VelvetVAE with other metabolic labelling datasets, supplementary to Figure 3. Plots showing genes per cell, UMIs per cell and first two principal components, coloured by either timepoint or UMIs per cell for A. sci-FATE, B. scNT-seq, C. well-TEMP-seq. All datasets have been filtered and quality-controlled. For all datasets, varying data quality (measured as UMIs per cell) provides a major source of confounding variation, and timecourse trajectories are not clearly evident. D. Comparison of VelvetVAE and Dynamo across these three datasets, quantified by cross-boundary correctness between the recorded timepoints.



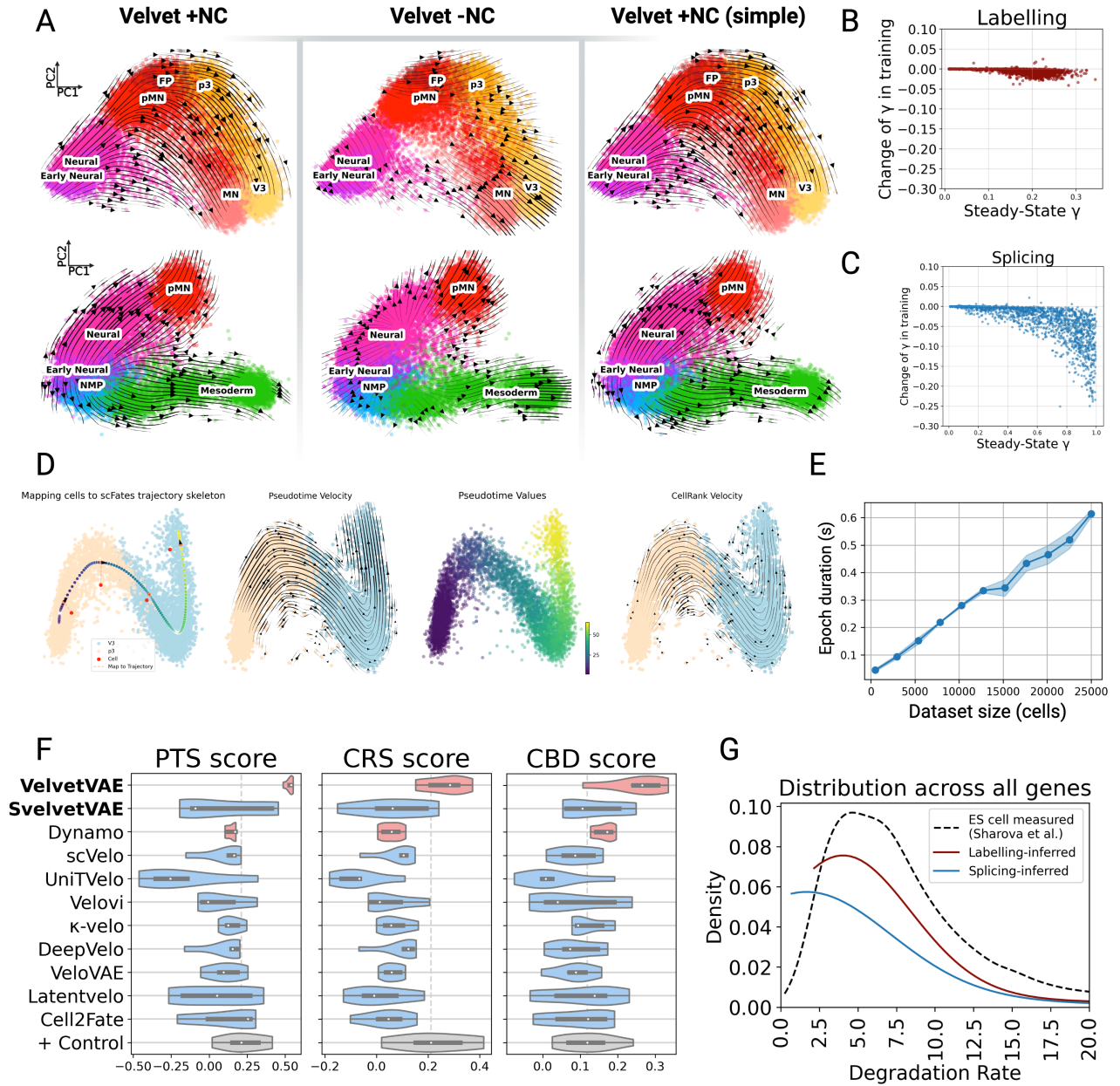
Supplementary Figure 6: Exploring velocity inference with SvelvetVAE and splicing data, supplementary to Figure 3. A. Benchmarking scores across publicly available splicing datasets comparing different velocity inference tools. Performance is assessed with cross-boundary directedness, with all tools generally scoring poorly. SvelvetVAE’s one negative score came from the gastrulation erythroid dataset [2], which has been known to cause reversed-direction velocity inference [3]. C. The best performing model instance, UniTVelo with dentate gyrus data [4], when visualised still performs poorly. D. Inferred velocities from the same dataset from SvelvetVAE with default parameters. E. A small adjustment of parameters, shifting the minimum value for γ from 0.1 to 0.01, reverses the inferred directionality in this instance, highlighting the lack of robustness in the temporal signal from splicing data.



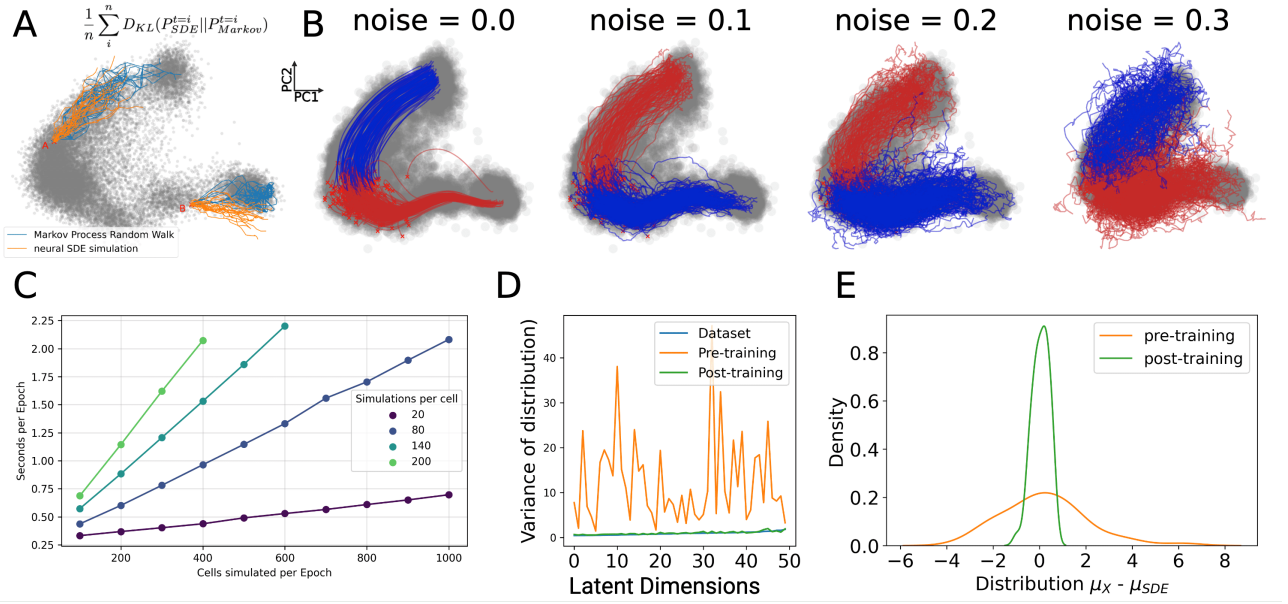
Supplementary Figure 7: Demonstration of Markov random walk simulations, supplementary to Figure 3. Simulating trajectories from early neural cells using random walks guided by a Markov transition matrix calculated from inferred velocities, as per [5] and others, produces noisy distributions of trajectories that do not easily cluster into the three expected fates, with cluster three composed of both MN and V3 fates. B. Simulating each cell's trajectory multiple times and examining the number of different fates predicted. C. Noisy gene expression profiles predicted from Markov random walk trajectories.



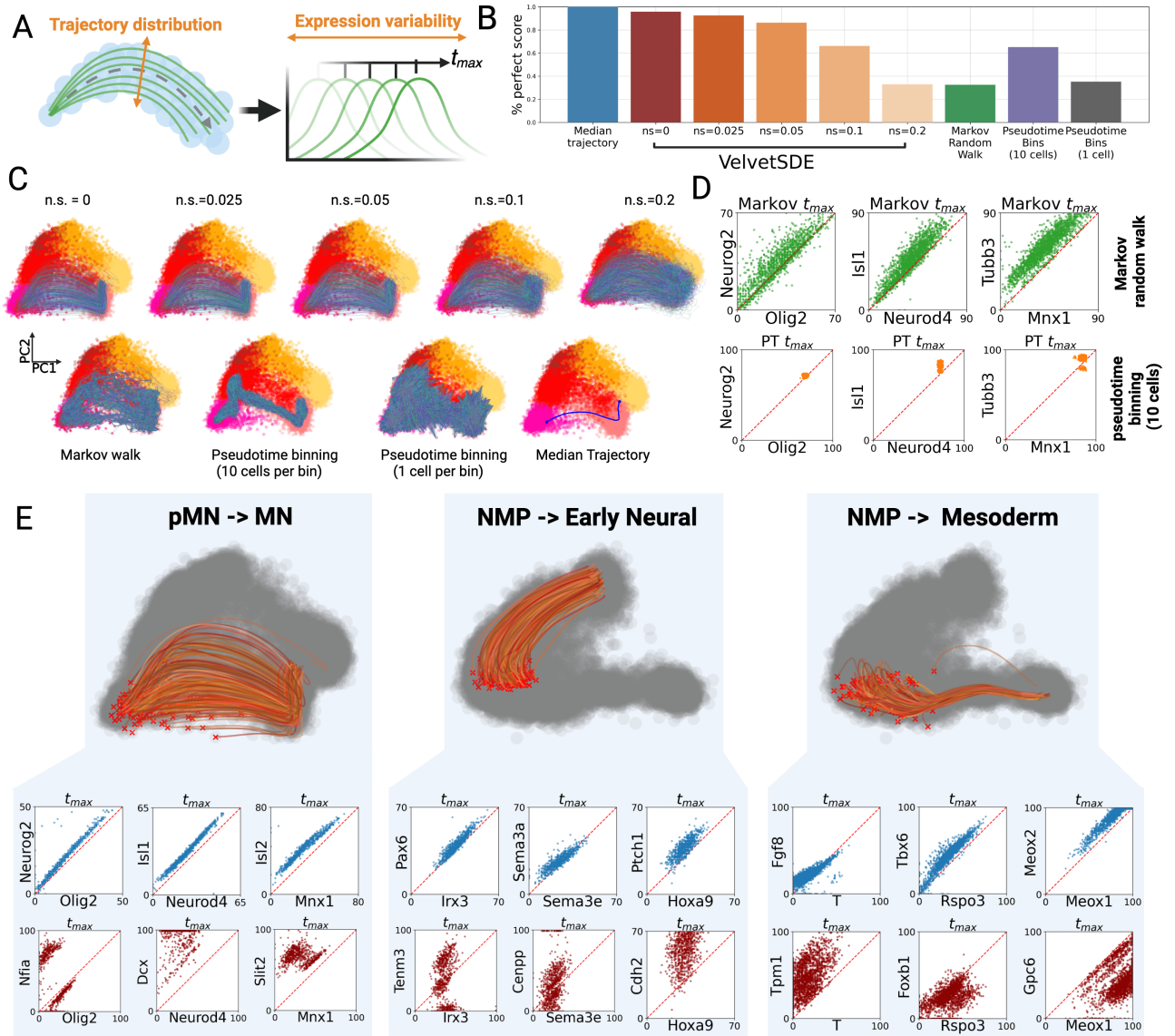
Supplementary Figure 8: Comparison of velocity dynamics from estimated vs. counted labelled reads from Dynast[6], supplementary to Figures 1 and 3 and supplementary note 1. A. Plotting the proportion of cells that contain both old (unlabelled) and new (labelled) reads for each gene, comparing estimated data versus counted data. Estimation leads to a consistent reduction in this proportion of data with both modalities observed, due to inflation of label rate. The most dynamically informative cells are ones that provide both labelled and unlabelled information. B and C. new (labelled versus old (unlabelled) reads for estimated data (B) and counted data (C). Estimation leads to a large increase in the number of cells with only new reads detected. D. Proportion of reads detected as 'new' (or labelled) when using counting or estimation. E. Distribution of correlation scores for each gene of label rate against total read count. Counting displays no correlation between label rate and total count, while with estimation, there is a consistent positive correlation. F. Comparing the inferred mRNA half-life distributions from counted and estimated data with experimentally measured half lives [7]. G. VelvetVAE velocity inference visualised with PCA of latent space (compare with velocity inference from counting data in Figure 3.) H and I. Dynamo velocity inference visualised with PCA, with counting data and estimation data, showing difference in inference is not VelvetVAE specific.



Supplementary Figure 9: Additional analysis of VelvetVAE, supplementary to Figure 3. A. Inferred velocity dynamics from VelvetVAE in two datasets (neural, top; NMP below), across three different settings: left, the default VelvetVAE model with neighbourhood constraint and a biophysical model of labelling. Middle, no neighbourhood constraint (NC). Right, with neighbourhood constraint, but a simplified model of labelling, where γ is just the expected steady-state proportion of total reads that are labelled. B and C. Plotting how learning γ as a learnable parameter of latent model affects the value: change of γ value is plotted against the original γ value for labelling-based and splicing-based inference. D. Schematic plot of constructing pseudotime ground-truths. Left; first, a pseudotime skeleton is constructed through manual parameter adjustment and selection of root nodes. For Pseudotime Score (PTS), each cell is given a velocity direction equal to the direction of the closest point in the skeleton to the subsequent point. Middle-left: example of the resultant PTS velocity vector field. Middle-right: From this skeleton, each value can also be assigned a pseudotime value, which can then be used by CellRank to construct a second velocity vector field, providing the ground truth for CellRank Score (CRS). Cross-boundary directedness (CBD) is unrelated to these ground truths. E. Training time per epoch against dataset size. Each additional 5000 cells increases training time by approximately 0.1 seconds per epoch. F. Benchmarking scores across pseudotime metrics with local averaging performed as a preprocessing step. Note: DeepVelo and VeloVAE are included in this analysis only as they require data to be locally averaged to function. Pyrovelocity is not included here as it does not function with locally-averaged data. In this context, local averaging refers to the preprocessing step where each cell-gene value is replaced with the average value of that cell's k nearest cells. G. Comparing the inferred mRNA half-life distributions from labelling and splicing data with experimentally measured half lives [7].



Supplementary Figure 10: Additional analysis of VelvetSDE, supplementary to Figure 3 and 4. A. Schematic demonstrating VelvetSDE’s training regime. Cell A shows close similarity between nSDE and Markov simulations, while cell B shows disagreement due to systematic error of nSDE. In training these distributions are compared, and the nSDE model is updated to such that the distributions better match. B. Visualisation of VelvetSDE trajectories with varying degrees of noise, used to empirically select noise magnitude for downstream analysis of fate boundaries. C. Training times for VelvetSDE, varied both in terms of number of simulated cells per epoch and number of simulations per cell (each for nSDE and Markov). On a single GPU core of a NVIDIA V100 GPU with 120Gb of memory, around 80,000 trajectories in total can be simulated before encountering memory issues - this still keeps a single epoch below 2 seconds in duration, and total training time in a few minutes. D. Per-dimension variance across latent dimensions, comparing original dataset (blue) with pre-training simulations (orange) and post-training VelvetSDE simulations (green). E. Distribution of per-dimension residuals (in other words, the mean of the dataset in a latent dimension minus the mean of the simulation) For pre-trained simulations (orange) and post-training VelvetSDE simulations (green). Together, D and E show how VelvetSDE training helps to constrain the variance of simulations and better replicate the distribution of the underlying dataset. F.



Supplementary Figure 12: Further analysis of trajectory distributions, supplementary to Figure 5. A. Schematic demonstrating the t_{max} scoring motivation: a distribution of trajectories can be simulated for a single fate. These trajectories can be mapped to gene expression space, giving a distribution of time series. The timepoint of maximal expression, t_{max} can be used to investigate correlative structure across this distribution for different genes. B. Full scoring of VelvetSDE trajectories across different noise magnitudes, scoring the percentage of trajectories that perfectly conserve the correct order of known sequential genes (for example, Olig2 should precede Neurog2). Performed for motor neuron trajectories. C. Visualisation of MN trajectories simulated via the different tested methods. D. Examining the distribution of t_{max} values for selected gene pairs from these pairs of sequential genes, we see that hidden Markov walks show noisy correlated distributions that sometimes show the correct order (above the red line) and sometimes not. Pseudotime binning removes this noisy correlative structure, but still leads to trajectories with incorrect orderings (below the red line). E. Extending this analysis with MN trajectories (left), NMP-to-neural trajectories (middle) and NMP-to-mesoderm trajectories (right), we find that related gene pairs (top, blue) often show correlations in the timing of their expression across trajectory distributions, but selected less related genes (bottom, red) show more complex or absent correlative structures, suggesting that there could be useful correlative information in distributions of trajectories that are lost when considering only a population average or pseudotime trajectory.

Parameter	Function	Default	Variation
Latent dimension (z)	Size of model's representation of data and vector field	50	Chosen empirically, based on complexity of a given dataset. Too small: multiple dynamics confounded on top of each other, inaccurate inference. Too large: sparse system makes dynamics difficult to learn
Neighbourhood space	Perform neighbourhood constraint in latent space of model or in full gene expression space	Latent space	In full gene space, cosine similarity suffers from curse of dimensionality and gradient calculations are extremely memory intensive. Calculating in latent space can also handle batch effects.
Biophysical model	Use explicit biophysical model of transcription and labelling, or simple new/old ratio model	Full biophysical model	With data used in this study, limited differences were observed, though it may be expected that the full biophysical model may be in general more robust.
Gamma mode	Whether to update degradation parameter in training or keep steady-state approximation	update	With data used in this study, updating gamma provides limited benefit to learning velocity dynamics, though may lead to more biophysically accurate gamma estimations
Velocity lambda	Magnitude of velocity penalty in training relative to VAE penalties	10	Too low and the vector field will not learn, too high and an appropriate mapping will not be learnt. A default of 10 is used to make the vector field loss more comparable in magnitude to VAE loss.
Training scheme	How many epochs of VAE training vs. neighbourhood constraint (NC) training	200 VAE epochs, 800 NC epochs	Training follows two stages: i) vector field + VAE, ii) vector field + NC. Training VAE and NC together can cause issues, as a cell's neighbouring cells in latent space is not fixed.
Trajectories per epoch	How many cells to simulate nSDE/Markov trajectories for per epoch	100	Should be adjusted with total number of epochs and training time. More trajectories per epoch more completely samples the dynamics of the system, but increases training time.
Simulations per trajectory	How many stochastic nSDE/Markov simulations to run per cell in each epoch	20	The more simulations per trajectories, the more comparable the nSDE and Markov simulations will be, however the slower training per epoch will be.
nSDE noise magnitude	The scalar magnitude of the random noise of nSDE	0.2	Should be empirically optimised such that nSDE simulation distributions show roughly similar distribution of trajectories and trajectory noise when compared to Markov walk simulations
Number of steps	How many forward steps to simulate with nSDE	50	Should be empirically optimised with number of markov steps, to ensure the two simulation strategies lead to trajectories of roughly same size
Number of markov steps	How many forward steps to simulate with Markov	10	Should be empirically optimised with number of steps, to ensure the two simulation strategies lead to trajectories of roughly same size
Tmax	Maximum time for nSDE simulations, determines how long trajectories should be.	10	Should be empirically optimised with number of steps and number of markov steps, but since the nSDE's time is relative, different tmax values will adjust the nSDE's relative "rapidity"
Beginning percentage	Only simulate trajectories initial percentage, based on precomputed pseudo-time values	not used (optional)	Useful if wanting to focus on the simulation of initial conditions, or if simulating later cells causes issues

Supplementary Figure 13: Table of key parameter and hyperparameters in VelvetVAE (blue) and VelvetSDE (green), and how inference might be affected by their variation.

Assumption	Assumption Details
Latent vector field	The intrinsic dimension of the dataset is lower than the dimensionality of observed features. This occurs when i) there are variables that add no meaningful variation, ii) when there are redundant or correlated variables. The use of a lower-dimensional latent vector field seeks to exploit the redundancies and correlative structures in the data to improve dynamical inference.
Continuous latent space	While single-cell RNA sequencing data is discrete (following a negative binomial distribution), the latent space of the model is continuous. This is because, as per scvi, we model the variation in the mean parameter for a negative binomial distribution for each gene. In this sense, the 'velocities' modelled can be thought of as time derivatives of these mean parameters.
Neighbourhood constraint	1) each cell's local neighbourhood is sufficiently well sampled. Where there are unobserved states, such as transitioning cells, this may be violated. 2) Velocities that point away from the convex hull of nearest neighbour cells are erroneous. 3) methods for calculating a cell's nearest neighbours can return other cells in the dataset that are similar to the current cell in a biological meaningful way (i.e. that 'similarity' is sufficiently determined by signal rather than noise).
Biophysical model	By using a biophysical model of labelling and transcription, we assume that metabolic labelling can sufficiently distinguish newly-transcribed molecules of RNA from pre-existing molecules. While a Bayesian approach is commonly used to improve this distinguishment, we find that it to be heavily distortive and do not use it.
Use of neural SDE	The dynamics of the system depend only on the current, observed state, and are modelled as a continuous-time process with fixed parameters over time.
Markov walk comparison	By using a Markov process in training of the nSDE, we assume that a Markovian approximation of dynamics is sufficiently accurate to update the nSDE vectorfield to keep trajectories in the data distribution without introducing unexpected or unobserved distortions. This assumption is derived from the above assumption that each cell's local neighbourhood is well sampled and that nearest neighbour similarity is sufficiently determined by biological signal rather than technical noise.
Use of scalar diagonal noise in training	Each latent dimension's noise intensity is equal and constant and independent from other dimensions, with Gaussian noise characteristics. The use of scalar diagonal noise is not a requirement but a modelling choice, and future work could look to extend modelling to more sophisticated forms of noise.

Supplementary Figure 14: Table of key modelling assumptions for VelvetVAE (blue) and VelvetSDE (green).